Lesson 22

NATIONAL INCOME DETERMINATION USING DIFFERENTIAL EQUATIONS

TOPIC 089: NATIONAL INCOME DETERMINATION USING DIFFERENTIAL EQUATIONS

If national income model is modelled by following system of equations:

$$Y = C + I$$

$$C = \alpha + \beta Y \qquad \alpha > 0,\ 0 < \beta < 1$$

α is the autonomous consumption and β is the marginal propensity to consume.

$$I = \delta Y + \gamma\left(\frac{dY}{dt}\right) \qquad \gamma > 0,\ \delta < 1$$

Find the time path and the intertemporal equilibrium level of national income.

Solution: Expenditure approach to National income is:

$$Y = \underline{C} + \underline{I}$$

Substituting values of C and I.

$$Y = \underline{\alpha + \beta Y} + \overline{\delta Y + \gamma\left(\frac{dY}{dt}\right)}$$

$$Y = \alpha + \beta Y + \delta Y + \gamma\left(\frac{dY}{dt}\right)$$

Rearranging to conform to a first order differential equation:

$$-\gamma\left(\frac{dY}{dt}\right) - \beta Y - \delta Y + Y = \alpha$$

$$-\gamma\left(\frac{dY}{dt}\right) - \beta Y - \delta Y + Y = \alpha$$

$$-\gamma\left(\frac{dY}{dt}\right) + Y - \beta Y - \delta Y = \alpha$$

$$-\gamma\left(\frac{dY}{dt}\right) + (1 - \beta - \delta)Y = \alpha$$

Dividing by $-\gamma$ throughout to normalize the differential equation.

$$\boxed{-\frac{\gamma}{(-\gamma)}}\left(\frac{dY}{dt}\right) \boxed{+} \frac{(1-\beta-\delta)}{(-\gamma)}Y = \frac{\alpha}{(-\gamma)}$$

$$\left(\frac{dY}{dt}\right) + \left(\frac{1-\beta-\delta}{-\gamma}\right)Y = \boxed{-}\frac{\alpha}{\gamma}$$

$$\left(\frac{dY}{dt}\right) + \left(\frac{1-\beta-\delta}{-\gamma}\right)Y = -\frac{\alpha}{\gamma}$$

Comparing with the standard form of first order differential equation; $\frac{dy}{dt} + ay = b$.

$$\underbrace{\left(\frac{dY}{dt}\right)}_{\frac{dy}{dt}} + \underbrace{\left(\frac{1-\beta-\delta}{-\gamma}\right)}_{a}Y = \underbrace{\left(-\frac{\alpha}{\gamma}\right)}_{b}$$

Here $\frac{dy}{dt} = \frac{dY}{dt}, y = Y, a = \left(\frac{1-\beta-\delta}{-\gamma}\right), b = \left(-\frac{\alpha}{\gamma}\right)$

Choice of Case of the Differential Equation:

$b = \left(-\frac{\alpha}{\gamma}\right)$; As $\alpha > 0$ and $\gamma > 0$ (both positive constants), their ratio will be non-zero positive constant. However, the value of $b = \left(-\frac{\alpha}{\gamma}\right)$ has a negative sign. The answer will be negative, though still non-zero. i.e. $b \neq 0$ **– Non-Homogenous Case**.

$\boldsymbol{a} = \left(\frac{1-\beta-\delta}{-\gamma}\right)$; Both $\boldsymbol{\beta}$ and $\boldsymbol{\delta}$ are less than $\mathbf{1}$. The numerator $(\mathbf{1}-\boldsymbol{\beta}-\boldsymbol{\delta})$ can become $\mathbf{0}$ only if $\boldsymbol{\beta}+\boldsymbol{\delta}=\mathbf{1}$. So, we assume that the $\boldsymbol{\beta}+\boldsymbol{\delta}\neq\mathbf{1}$ and $(\mathbf{1}-\boldsymbol{\beta}-\boldsymbol{\delta}\neq\mathbf{0})$, so $\left(\frac{1-\beta-\delta}{-\gamma}\neq 0\right)$ i.e. $\boldsymbol{a}\neq\mathbf{0}$ **– Non-Homogenous Case with $a \neq 0$.**

Therefore, using the definite Solution of time-path of $NHC_{a\neq 0}$:

$$y(t) = \left\{y(0) - \frac{b}{a}\right\} e^{-at} + \frac{b}{a}$$

Emphasizing on the values to be substituted:

$$\underline{y}(t) = \left\{\underline{y(0)} - \frac{\boxed{b}}{\boxed{a}}\right\} e^{-\boxed{a}t} + \frac{\boxed{b}}{\boxed{a}}$$

Substituting values of y, a and b, respectively:

$$\underline{Y}(t) = \left\{\underline{Y(0)} - \frac{\boxed{\left(-\frac{\alpha}{\gamma}\right)}}{\boxed{\left(\frac{1-\beta-\delta}{-\gamma}\right)}}\right\} e^{-\boxed{\left(\frac{1-\beta-\delta}{-\gamma}\right)}t} + \frac{\boxed{\left(-\frac{\alpha}{\gamma}\right)}}{\boxed{\left(\frac{1-\beta-\delta}{-\gamma}\right)}}$$

$$Y(t) = \left\{Y(0) - \frac{\left(\boxed{-}\frac{\alpha}{\gamma}\right)}{\left(\frac{1-\beta-\delta}{\boxed{-}\gamma}\right)}\right\} e^{\boxed{-}\left(\frac{1-\beta-\delta}{\boxed{-}\gamma}\right)t} + \frac{\left(\boxed{-}\frac{\alpha}{\gamma}\right)}{\left(\frac{1-\beta-\delta}{\boxed{-}\gamma}\right)}$$

$$Y(t) = \left\{Y(0) - \frac{(\alpha)}{(1-\beta-\delta)}\right\} e^{\left(\frac{1-\beta-\delta}{\gamma}\right)t} + \frac{(\alpha)}{(1-\beta-\delta)}$$

$$\boxed{Y(t) = \left\{Y(0) - \left(\frac{\alpha}{1-\beta-\delta}\right)\right\} e^{\left(\frac{1-\beta-\delta}{\gamma}\right)t} + \left(\frac{\alpha}{1-\beta-\delta}\right)}$$

Definite Time-path for National Income in Dynamic Sense.

Dynamic Stability: For the dynamic stability of the time-path of First Order Differential Equation, it should converge towards equilibrium. We know that equilibrium is particular solution (Y_p). Whereas convergence happens when deviation from equilibrium fades to zero. In a differential equation, deviation component is measured by (Y_c). Therefore, for dynamic stability; $Y_c \to 0$ as $t \to \infty$.

Using the Definite Time-path for Income level in Dynamic Sense.

$$Y(t) = \underbrace{\left\{Y(0) - \left(\frac{\alpha}{1-\beta-\delta}\right)\right\} e^{\left(\frac{1-\beta-\delta}{\gamma}\right)t}}_{Y_c} + \underbrace{\left(\frac{\alpha}{1-\beta-\delta}\right)}_{Y_p}$$

Need for time path of income $\{Y(t)\}$ to converge to equilibrium income (Y^*) i.e. $Y(t) \to Y^*$ as $Y_c \to 0$ when $t \to \infty$.

$$Y_c \to 0$$

$$\left\{Y(0) - \left(\frac{\alpha}{1-\beta-\delta}\right)\right\} e^{\left(\frac{1-\beta-\delta}{\gamma}\right)t} \to 0$$

However, on hair splitting, we don't get the desired result.

$$\left[\underbrace{\left\{\underbrace{Y(0) - \overbrace{\boxed{\left(\frac{\alpha}{1-\beta-\delta}\right)}}^{\{\alpha,\beta,\delta>0\}}}_{Y(0)\gtreqless Y_p}\right\}}_{Deviation\ (+ve\ or -ve)} \underbrace{e^{\overbrace{\boxed{\left(\frac{1-\beta-\delta}{\gamma}\right)}}^{\{\beta,\delta,\gamma>0\}}t}}_{\substack{Exponential\\ Growth\\ over\ time}}\right] \to 0$$

Therefore,

$$Y(t) = \underbrace{\left\{Y(0) - \left(\frac{\alpha}{1-\beta-\delta}\right)\right\} e^{\left(\frac{1-\beta-\delta}{\gamma}\right)t}}_{\to\infty} + \left(\frac{\alpha}{1-\beta-\delta}\right)$$

$$Y(t) = \underbrace{\left\{Y(0) - \boxed{\left(\frac{\alpha}{1-\beta-\delta}\right)}\right\} e^{\left(\frac{1-\beta-\delta}{\gamma}\right)t}}_{\to\infty} + \boxed{\left(\frac{\alpha}{1-\beta-\delta}\right)}$$

$$Y(t) = \underbrace{\left\{Y(0) - \boxed{Y_p}\right\} e^{\left(\frac{1-\beta-\delta}{\gamma}\right)t}}_{\to\infty} + \boxed{Y_p}$$

As particular integral is also the equilibrium value of the variable $\boxed{Y_p = Y^*}$.

$$Y(t) = \underbrace{\left\{Y(0) - \boxed{Y^*}\right\} e^{\left(\frac{1-\beta-\delta}{\gamma}\right)t}}_{\to\infty} + \boxed{Y^*}$$

Since addition of (Y^*) in ∞ shall have no effect on infinity;

$$\boxed{Y(t) \to \infty}$$

$$\boxed{Y(t) \nrightarrow Y^*}$$

Interpretation: It can be said that provided $\beta > 0, \delta < 1$ and $\gamma > 0$, regardless of initial national income $Y(0)$ is greater or smaller than equilibrium national income (Y^*), the time-path of national income shall **<u>diverge away from equilibrium</u>** level of national income as time is allowed to pass indefinitely. Hence the time-path is **<u>dynamically unstable</u>**.

<u>TOPIC 090: NATIONAL INCOME DETERMINATION: NUMERICAL</u>

Considering a two-sector economy; we have following structural equations:

$$\frac{dY}{dt} = 0.5(C + I - Y)$$

$$C = 0.8Y + 400$$

$$I = 600$$

Find the time-path of national income of the economy; $Y(0) = 7000$.

Solution: Given the equation for rate of national income $\frac{dY}{dt} = 0.5(C + I - Y)$; we can substitute equations of C and I.

$$\frac{dY}{dt} = 0.5(\underline{C} + \underline{I} - Y)$$

$$\frac{dY}{dt} = 0.5(\underline{0.8Y + 400} + \underline{600} - Y)$$

$$\frac{dY}{dt} = 0.5(0.8Y - Y + 400 + 600)$$

$$\frac{dY}{dt} = 0.5(-0.2Y + 1000)$$

$$\frac{dY}{dt} = -0.1Y + 500$$

$$\boxed{\frac{dY}{dt} + 0.1Y = 500}$$

First Order Differential Equation for National Income.

$$\underbrace{\frac{dY}{dt}}_{\frac{dy}{dt}} + \underbrace{0.1}_{a} Y = \underbrace{500}_{b}$$

Here $\frac{dy}{dt} = \frac{dY}{dt}$, $\boldsymbol{y = Y}$, $\boldsymbol{a = 0.1}$, $\boldsymbol{b = 500}$, $\boldsymbol{y(0) = Y(0) = 7000}$

Choice of Case of the Differential Equation:

As $\boldsymbol{b = 500}$, so $\boldsymbol{b \neq 0}$**:** Non-Homogenous Case.

As $\boldsymbol{a = 0.1}$, so $\boldsymbol{a \neq 0}$**:** Non-Homogenous Case with $\boldsymbol{a \neq 0}$.

Therefore, using the definite Solution of time-path of $\boldsymbol{NHC_{a \neq 0}}$:

$$\boldsymbol{y(t) = \left\{y(0) - \frac{b}{a}\right\} e^{-at} + \frac{b}{a}}$$

Emphasizing on the values to be substituted:

$$\boldsymbol{\underline{y}(t) = \left\{\underline{y(0)} - \frac{\boxed{b}}{\boxed{a}}\right\} e^{-\boxed{a}t} + \frac{\boxed{b}}{\boxed{a}}}$$

Substituting values of $\boldsymbol{y}$, $\boldsymbol{a}$**,** $\boldsymbol{b}$ and $\boldsymbol{y(0)}$ respectively:

$$\boldsymbol{\underline{Y}(t) = \left(\underline{Y(0)} - \frac{\boxed{500}}{\boxed{0.1}}\right) e^{-\boxed{0.1}t} + \frac{\boxed{500}}{\boxed{0.1}}}$$

$$\boldsymbol{Y(t) = \left(\underline{7000} - \frac{500}{0.1}\right) e^{-0.1t} + \frac{500}{0.1}}$$

$$\boldsymbol{Y(t) = (7000 - 5000)e^{-0.1t} + 5000}$$

$$\boldsymbol{Y(t) = (2000)e^{-0.1t} + 5000}$$

$$\boxed{\boldsymbol{Y(t) = 2000\, e^{-0.1t} + 5000}}$$

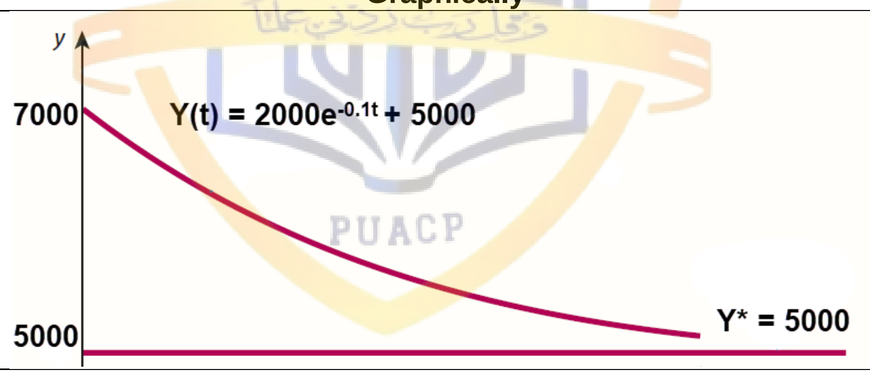
It is the definite Time-Path of National Income of the economy.

$$\boldsymbol{Y(t) = \underbrace{2000e^{-0.1t}}_{Y_c} + \underbrace{5000}_{Y_p}}$$

Dynamic Stability: The expression; $\boldsymbol{Y(t) = \boxed{2000e^{-0.1t}} + 5000}$ has an exponential decay term in the box. **i.e.**:

t	Complementary Function (Deviation) $Y_c = 2000e^{-0.1t}$	Time-Path $Y(t) = 2000e^{-0.1t} + 5000$
0	2000	7000
1	1809.675	6809.674836
20	270.6706	5270.670566
40	36.63128	5036.631278
60	4.957504	5004.957504
80	0.670925	5000.670925
100	0.090800	5000.090800
120	0.012288	5000.012288
140	0.001663	5000.001663
160	0.000225	5000.000225
200	0.000004	5000.000004
…	…	…

	Remarks	
	Positive deviation decays over time	Time-path **slides and converges** to equilibrium ($Y^* = 5000$)
	As $t \to \infty$, $Y_c \to 0$	And when $Y_c \to 0$, $(Y_t)\overrightarrow{Converges}(Y^*)$
	Graphically	



The straight line shows the equilibrium value of national income (Y^*), while curve shows the deviation from equilibrium (Y_c), which decreases over time and leads the time-path to convergence.

TOPIC 091: INCOME DYNAMICS & INDUCED GOVERNMENT SPENDING USING DIFFERENTIAL EQUATIONS

If national income model is modelled by following system of equations:

$$Y = C + I + G$$

$$C = \alpha + \beta Y \qquad \alpha > 0,\ 0 < \beta$$

α is the autonomous consumption and β is the marginal propensity to consume.
Following is the investment function:

$$I = \delta Y + \gamma\left(\frac{dY}{dt}\right) \qquad \gamma > 0,\ \delta < 1$$

Government Spending is included as the 3rd sector:

$$G = gY \qquad g < 1$$

$\boldsymbol{g}$ is the proportion of national income spent as government expenditure.
Find the time path and the intertemporal equilibrium level of national income.
Solution: Expenditure approach to National income is:

$$\boldsymbol{Y = \underline{C} + \underline{I} + \underline{G}}$$

Substituting values of $\boldsymbol{C}$, $\boldsymbol{I}$ and $\boldsymbol{G}$.

$$\boldsymbol{Y = \underline{\alpha + \beta Y} + \underline{\delta Y + \gamma\left(\frac{dY}{dt}\right)} + \underline{gY}}$$

$$\boldsymbol{Y = \alpha + \beta Y + \delta Y + \gamma\left(\frac{dY}{dt}\right) + gY}$$

Rearranging to conform to a first order differential equation:

$$\boldsymbol{-\gamma\left(\frac{dY}{dt}\right) + Y - \beta Y - \delta Y - gY = \alpha}$$

$$\boldsymbol{-\gamma\left(\frac{dY}{dt}\right) + (1 - \beta - \delta - g)Y = \alpha}$$

Dividing by $\boldsymbol{-\gamma}$ throughout to normalize the differential equation.

$$\boxed{-\frac{\boldsymbol{\gamma}}{(-\boldsymbol{\gamma})}}\left(\frac{\boldsymbol{dY}}{\boldsymbol{dt}}\right)\boxed{+}\frac{(\boldsymbol{1 - \beta - \delta - g})}{(-\boldsymbol{\gamma})}\boldsymbol{Y} = \frac{\boldsymbol{\alpha}}{(-\boldsymbol{\gamma})}$$

$$\boldsymbol{\left(\frac{dY}{dt}\right) + \left(\frac{1 - \beta - \delta - g}{-\gamma}\right)Y = \boxed{-}\frac{\alpha}{\gamma}}$$

$$\boldsymbol{\left(\frac{dY}{dt}\right) + \left(\frac{1 - \beta - \delta - g}{-\gamma}\right)Y = -\frac{\alpha}{\gamma}}$$

Comparing with the standard form of first order differential equation; $\boldsymbol{\frac{dy}{dt} + ay = b}$.

$$\boldsymbol{\underbrace{\left(\frac{dY}{dt}\right)}_{\frac{dy}{dt}} + \underbrace{\left(\frac{1 - \beta - \delta - g}{-\gamma}\right)}_{a}Y = \underbrace{\left(-\frac{\alpha}{\gamma}\right)}_{b}}$$

Here $\boldsymbol{\frac{dy}{dt} = \frac{dY}{dt}, y = Y, a = \left(\frac{1-\beta-\delta-g}{-\gamma}\right), b = \left(-\frac{\alpha}{\gamma}\right)}$

Choice of Case of the Differential Equation:

$\boldsymbol{b = \left(-\frac{\alpha}{\gamma}\right)}$; As $\boldsymbol{\alpha > 0}$ and $\boldsymbol{\gamma > 0}$ (both positive constants), their ratio will be non-zero positive constant. However, the value of $\boldsymbol{b = \left(-\frac{\alpha}{\gamma}\right)}$ has a negative sign. The answer will be negative, though still non-zero. i.e. $\boldsymbol{b \neq 0}$ **– Non-Homogenous Case**.

$\boldsymbol{a = \left(\frac{1-\beta-\delta-g}{-\gamma}\right)}$; the numerator $(\boldsymbol{1 - \beta - \delta - g})$ can be non-zero if $(\boldsymbol{\beta + \delta + g}) \neq \boldsymbol{1}$. Also $\boldsymbol{\gamma}$ is positive value. Therefore, value of $\boldsymbol{a}$ will be some positive/negative (non-zero) value. i.e. $\boldsymbol{a \neq 0}$ **– Non-Homogenous Case with $\boldsymbol{a \neq 0}$.**

Therefore, using the definite Solution of time-path of $\boldsymbol{NHC_{a\neq 0}}$:

$$\boldsymbol{y(t) = \left\{y(0) - \frac{b}{a}\right\}e^{-at} + \frac{b}{a}}$$

Emphasizing on the values to be substituted:

$$\boldsymbol{\underline{y}(t) = \left\{\underline{y(0)} - \frac{\boxed{b}}{\boxed{a}}\right\}e^{-\boxed{a}t} + \frac{\boxed{b}}{\boxed{a}}}$$

Substituting values of $\boldsymbol{y}$, $\boldsymbol{a}$ and $\boldsymbol{b}$, respectively:

$$\boldsymbol{\underline{Y}(t) = \left\{\underline{Y(0)} - \frac{\boxed{\left(-\frac{\alpha}{\gamma}\right)}}{\boxed{\left(\frac{1 - \beta - \delta - g}{-\gamma}\right)}}\right\}e^{-\boxed{\left(\frac{1-\beta-\delta-g}{-\gamma}\right)}t} + \frac{\boxed{\left(-\frac{\alpha}{\gamma}\right)}}{\boxed{\left(\frac{1 - \beta - \delta - g}{-\gamma}\right)}}}$$

$$Y(t) = \left\{Y(0) - \frac{\left(\square \frac{\alpha}{\gamma}\right)}{\left(\frac{1-\beta-\delta-g}{\square \gamma}\right)}\right\} e^{\square\left(\frac{1-\beta-\delta-g}{\square \gamma}\right)t} + \frac{\left(\square \frac{\alpha}{\gamma}\right)}{\left(\frac{1-\beta-\delta-g}{\square \gamma}\right)}$$

$$Y(t) = \left\{Y(0) - \frac{(\alpha)}{(1-\beta-\delta-g)}\right\} e^{\left(\frac{1-\beta-\delta-g}{\gamma}\right)t} + \frac{(\alpha)}{(1-\beta-\delta-g)}$$

$$\boxed{Y(t) = \left\{Y(0) - \left(\frac{\alpha}{1-\beta-\delta-g}\right)\right\} e^{\left(\frac{1-\beta-\delta-g}{\gamma}\right)t} + \left(\frac{\alpha}{1-\beta-\delta-g}\right)}$$

Definite Time-path for National Income in Dynamic Sense. And g has positive effect on the national income, via its negative sign in denominator.

Dynamic Stability: For the dynamic stability of the time-path of First Order Differential Equation, it should converge towards equilibrium. We know that equilibrium is particular solution (Y_p). Whereas convergence happens when deviation from equilibrium fades to zero. In a differential equation, deviation component is measured by (Y_c). Therefore, for dynamic stability; $Y_c \to 0$ as $t \to \infty$.

Using the Definite Time-path for Income level in Dynamic Sense.

$$Y(t) = \underbrace{\left\{Y(0) - \left(\frac{\alpha}{1-\beta-\delta-g}\right)\right\} e^{\left(\frac{1-\beta-\delta-g}{\gamma}\right)t}}_{Y_c} + \underbrace{\left(\frac{\alpha}{1-\beta-\delta-g}\right)}_{Y_p}$$

Need for time path of income $\{Y(t)\}$ to converge to equilibrium income (Y^*) i.e. $Y(t) \to Y^*$ as $Y_c \to 0$ when $t \to \infty$.

$$Y_c \to 0$$

$$\left\{Y(0) - \left(\frac{\alpha}{1-\beta-\delta-g}\right)\right\} e^{\left(\frac{1-\beta-\delta-g}{\gamma}\right)t} \to 0$$

However, on hair splitting, we don't get the desired result.

$$\left[\underbrace{\left\{\underbrace{Y(0) - \overbrace{\boxed{\left(\frac{\alpha}{1-\beta-\delta-g}\right)}}^{\{\alpha,\beta,\delta,g>0\}}}_{Y(0)\gtrless Y_p}\right\}}_{Deviation\ (+ve\ or -ve)} \underbrace{e^{\overbrace{\boxed{\left(\frac{1-\beta-\delta-g}{\gamma}\right)}}^{\{\beta,\delta,\gamma,g>0\}} t}}_{\substack{Exponential\\ Growth\\ over\ time}}\right] \to 0$$

Therefore,

$$Y(t) = \underbrace{\left\{Y(0) - \left(\frac{\alpha}{1-\beta-\delta-g}\right)\right\} e^{\left(\frac{1-\beta-\delta-g}{\gamma}\right)t}}_{\to\infty} + \left(\frac{\alpha}{1-\beta-\delta-g}\right)$$

$$Y(t) = \underbrace{\left\{Y(0) - \boxed{\left(\frac{\alpha}{1-\beta-\delta-g}\right)}\right\} e^{\left(\frac{1-\beta-\delta-g}{\gamma}\right)t}}_{\to\infty} + \boxed{\left(\frac{\alpha}{1-\beta-\delta-g}\right)}$$

$$Y(t) = \underbrace{\left\{Y(0) - \boxed{Y_p}\right\} e^{\left(\frac{1-\beta-\delta-g}{\gamma}\right)t}}_{\to\infty} + \boxed{Y_p}$$

As particular integral is also the equilibrium value of the variable $\boxed{Y_p = Y^*}$.

$$Y(t) = \underbrace{\left\{Y(0) - \boxed{Y^*}\right\} e^{\left(\frac{1-\beta-\delta-g}{\gamma}\right)t}}_{\to\infty} + \boxed{Y^*}$$

Since addition of (Y^*) in ∞ shall have no effect on infinity;

$$\boxed{Y(t) \to \infty}$$

$$\boxed{Y(t) \nrightarrow Y^*}$$

Interpretation: It can be said that provided $\beta > 0, \delta < 1, \gamma > 0$ and $g < 1$, regardless of initial national income $Y(0)$ is greater or smaller than equilibrium national income (Y^*), the time-path of national income shall **diverge away from equilibrium** level of national income as time is allowed to pass indefinitely. Hence the time-path is **dynamically unstable**.

TOPIC 092: DYNAMICS OF INVESTMENT USING DIFFERENTIAL EQUATIONS

A principal of $\$60$ is invested. The value, $I(t)$, of the investment, t days later, satisfies the differential equation $\frac{dI}{dt} = 0.002I + 5$. Find the value of the investment if the initial level of investment $y(0)$ is 2000.

Solution: Given the equation for rate of investment $\frac{dI}{dt} = 0.002I + 5$; we can rearrange to compare with the standard form of differential equation.

$$\frac{dI}{dt} = 0.002I + 5$$

$$\frac{dI}{dt} - 0.002I = 5$$

$$\underbrace{\frac{dI}{dt}}_{\frac{dy}{dt}} + \underbrace{(0.002)}_{a} I = \underbrace{5}_{b}$$

Here $\frac{dy}{dt} = \frac{dI}{dt}$, $y = I$, $a = 0.002$, $b = 5$, $y(0) = I(0) = 2000$.

Choice of Case of the Differential Equation:

As $b = 5$, so $b \neq 0$**:** Non-Homogenous Case.

As $a = 0.002$, so $a \neq 0$**:** Non-Homogenous Case with $a \neq 0$.

Therefore, using the Definite Solution of time-path of $NHC_{a\neq 0}$:

$$y(t) = \left\{y(0) - \frac{b}{a}\right\} e^{-at} + \frac{b}{a}$$

Emphasizing on the values to be substituted:

$$\underline{y}(t) = \left\{\underline{y(0)} - \frac{\boxed{b}}{\boxed{a}}\right\} e^{-\boxed{a}t} + \frac{\boxed{b}}{\boxed{a}}$$

Substituting values of y, a, b and $y(0)$ respectively:

$$\underline{I}(t) = \left(\underline{I(0)} - \frac{\boxed{5}}{\boxed{0.002}}\right) e^{-\boxed{0.002}t} + \frac{\boxed{5}}{\boxed{0.002}}$$

$$I(t) = \left(\underline{2000} - \frac{5}{0.002}\right) e^{-0.002t} + \frac{5}{0.002}$$

$$I(t) = (2000 - 2500)e^{-0.002t} + 2500$$

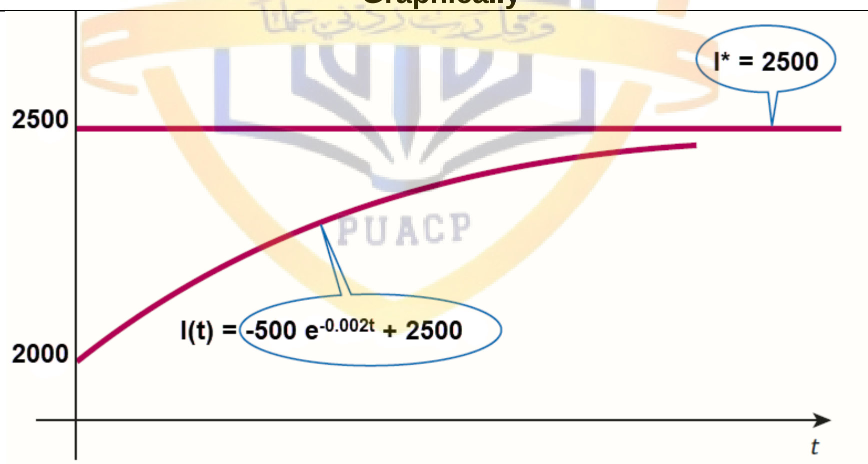
$$\boxed{I(t) = -500e^{-0.002t} + 2500}$$

It is the definite Time-Path of Investment.

$$I(t) = \underbrace{-500e^{-0.002t}}_{I_c} + \underbrace{2500}_{I_p}$$

Dynamic Stability: The expression; $I(t) = \boxed{-500e^{-0.002t}} + 2500$ has an exponential decay term in the box. **i.e.**:

t	Complementary Function (Deviation) $I_c = -500e^{-0.002t}$	Time-Path $I(t) = -500e^{-0.002t} + 2500$
0	-500.0000	2000.0000
1	-499.0009	2000.9990
2	-498.0040	2001.9960
3	-497.0089	2002.9910
4	-496.0160	2003.9840
5	-495.0249	2004.9751
6	-494.0359	2005.9641
7	-493.0488	2006.9512
8	-492.0637	2007.9363
9	-491.0805	2008.9194
10	-490.0993	2009.9006
…	…	…

	Remarks	
	Negative deviation decays over time	Time-path **rises and converges** to equilibrium $(I^* = 2500)$
	As $t \to \infty$, $I_c \to 0$	And when $I_c \to 0$, $(I_t)\overrightarrow{Converges}(I^*)$
	Graphically	
		
	The straight line shows the equilibrium value of consumption (I^*), while curve shows the deviation from equilibrium (I_c), and this deviation decreases over time and leads the time-path to convergence.	

TOPIC 093: DYNAMICS OF INTEREST RATE IN CAPITAL MARKET USING DIFFERENTIAL EQUATIONS

The demand and supply for borrowed capital in the capital market are given by the investment and savings functions, respectively:

$$I = \alpha - \beta i + \rho \frac{di}{dt} \qquad S = -\gamma + \delta i + \sigma \frac{di}{dt} \qquad (\alpha, \beta, \gamma, \delta)$$

Determine the intertemporal equilibrium level and time path of interest rate $i(t)$. What values of the parameters ρ and σ ensure dynamic stability?

Solution: Equilibrium in the market requires equality of Investment and Savings: ($\boldsymbol{S = I}$).

$$\boldsymbol{\alpha - \beta i + \rho \frac{di}{dt} = -\gamma + \delta i + \sigma \frac{di}{dt}}$$

Rearranging to compare with the standard form of differential equation.

$$\boldsymbol{-\beta i - \delta i + \rho \frac{di}{dt} - \sigma \frac{di}{dt} = -\alpha - \gamma}$$

$$\boldsymbol{\rho \frac{di}{dt} - \sigma \frac{di}{dt} - \beta i - \delta i = -\alpha - \gamma}$$

$$\boldsymbol{(\rho - \sigma) \frac{di}{dt} - (\beta + \delta) i = -(\alpha + \gamma)}$$

Dividing by $(\boldsymbol{\rho - \sigma})$ for normalizing the differential equation.

$$\boldsymbol{\frac{\cancel{(\rho - \sigma)}}{\cancel{(\rho - \sigma)}} \frac{di}{dt} - \frac{(\beta + \delta)}{(\rho - \sigma)} i = -\frac{(\alpha + \gamma)}{(\rho - \sigma)}}$$

$$\boldsymbol{\frac{di}{dt} - \frac{(\beta + \delta)}{(\rho - \sigma)} i = -\frac{(\alpha + \gamma)}{(\rho - \sigma)}}$$

Comparing with the standard form of the differential equation, $\boldsymbol{\frac{dy}{dt} + ay = b}$.

$$\underbrace{\boldsymbol{\frac{di}{dt}}}_{\frac{dy}{dt}} + \underbrace{\boldsymbol{\left\{-\frac{(\beta + \delta)}{(\rho - \sigma)}\right\}}}_{a} \underbrace{\boldsymbol{i}}_{y} = \underbrace{\boldsymbol{-\frac{(\alpha + \gamma)}{(\rho - \sigma)}}}_{b}$$

Here $\boldsymbol{\frac{dy}{dt} = \frac{di}{dt}, y = i, a = \left\{-\frac{(\beta+\delta)}{(\rho-\sigma)}\right\}, b = \left\{-\frac{(\alpha+\gamma)}{(\rho-\sigma)}\right\}}$

Choice of Case of the Differential Equation:

$\boldsymbol{b} = \left\{-\frac{(\alpha+\gamma)}{(\rho-\sigma)}\right\}$; As $\boldsymbol{\alpha > 0}$ and $\boldsymbol{\gamma > 0}$ (both positive constants), their ratio will be non-zero positive constant. However, the value of $\boldsymbol{b} = \left\{-\frac{(\alpha+\gamma)}{(\rho-\sigma)}\right\}$ who's denominator should be non-zero to avoid undefined answer (∞). The answer will be negative, though still non-zero. i.e. $\boldsymbol{b \neq 0}$ **– Non-Homogenous Case**.

$\boldsymbol{a} = \left\{-\frac{(\beta+\delta)}{(\rho-\sigma)}\right\}$; Both $\boldsymbol{\beta}$ and $\boldsymbol{\delta}$ are less than $\mathbf{1}$. The numerator $(\boldsymbol{\beta + \delta})$ shall be a non-zero positive constant. However, $\boldsymbol{a} = \left\{-\frac{(\beta+\delta)}{(\rho-\sigma)}\right\}$ so value of numerator $(\boldsymbol{\rho - \sigma})$ should be non-zero to avoid undefined answer (∞). So $\left[\left\{-\frac{(\beta+\delta)}{(\rho-\sigma)}\right\} \neq \mathbf{0}\right]$ i.e. $\boldsymbol{a \neq 0}$ **– Non-Homogenous Case with $\boldsymbol{a \neq 0}$.**

Therefore, using the definite Solution of time-path of $\boldsymbol{NHC_{a\neq 0}}$:

$$\boldsymbol{y(t) = \left\{y(0) - \frac{b}{a}\right\} e^{-at} + \frac{b}{a}}$$

Emphasizing on the values to be substituted:

$$\boldsymbol{\underline{y}(t) = \left\{\underline{y(0)} - \frac{\boxed{b}}{\boxed{a}}\right\} e^{-\boxed{a}t} + \frac{\boxed{b}}{\boxed{a}}}$$

Substituting values of $\boldsymbol{y}$, $\boldsymbol{a}$ and $\boldsymbol{b}$, respectively:

$$\boldsymbol{\underline{i}(t) = \left\{\underline{i(0)} - \frac{\boxed{\left\{-\frac{(\alpha+\gamma)}{(\rho-\sigma)}\right\}}}{\boxed{\left\{-\frac{(\beta+\delta)}{(\rho-\sigma)}\right\}}}\right\} e^{-\boxed{\left\{-\frac{(\beta+\delta)}{(\rho-\sigma)}\right\}}t} + \frac{\boxed{\left\{-\frac{(\alpha+\gamma)}{(\rho-\sigma)}\right\}}}{\boxed{\left\{-\frac{(\beta+\delta)}{(\rho-\sigma)}\right\}}}}$$

$$\boldsymbol{\underline{i}(t) = \left[\underline{i(0)} - \frac{\left\{\boxminus\frac{(\alpha+\gamma)}{(\rho-\sigma)}\right\}}{\left\{\boxminus\frac{(\beta+\delta)}{(\rho-\sigma)}\right\}}\right] e^{\boxminus\left\{\boxminus\frac{(\beta+\delta)}{(\rho-\sigma)}\right\}t} + \frac{\left\{\boxminus\frac{(\alpha+\gamma)}{(\rho-\sigma)}\right\}}{\left\{\boxminus\frac{(\beta+\delta)}{(\rho-\sigma)}\right\}}}$$

$$i(t)=\left[i(0)-\frac{\left\{\frac{((\alpha+\gamma))}{\cancel{(\rho-\sigma)}}\right\}}{\left\{\frac{((\beta+\delta))}{\cancel{(\rho-\sigma)}}\right\}}\right]e^{\left(\frac{\beta+\delta}{\rho-\sigma}\right)t}+\frac{\left\{\frac{((\alpha+\gamma))}{\cancel{(\rho-\sigma)}}\right\}}{\left\{\frac{((\beta+\delta))}{\cancel{(\rho-\sigma)}}\right\}}$$

$$i(t)=\left\{i(0)-\left(\frac{\alpha+\gamma}{\beta+\delta}\right)\right\}e^{\left(\frac{\beta+\delta}{\rho-\sigma}\right)t}+\left(\frac{\alpha+\gamma}{\beta+\delta}\right)$$

$$\boxed{i(t)=\left\{i(0)-\left(\frac{\alpha+\gamma}{\beta+\delta}\right)\right\}e^{\left(\frac{\beta+\delta}{\rho-\sigma}\right)t}+\left(\frac{\alpha+\gamma}{\beta+\delta}\right)}$$

Definite Time-path for Interest rate in Dynamic Sense.

Dynamic Stability: For the dynamic stability of the time-path of First Order Differential Equation, it should converge towards equilibrium. We know that equilibrium is particular solution (i_p). Whereas convergence happens when deviation from equilibrium fades to zero. In a differential equation, deviation component is measured by (i_c). Therefore, for dynamic stability; $i_c \to 0$ as $t \to \infty$.

Using the Definite Time-path for interest rate in Dynamic Sense.

$$i(t)=\underbrace{\left\{i(0)-\left(\frac{\alpha+\gamma}{\beta+\delta}\right)\right\}e^{\left(\frac{\beta+\delta}{\rho-\sigma}\right)t}}_{i_c}+\underbrace{\left(\frac{\alpha+\gamma}{\beta+\delta}\right)}_{i_p}$$

Need for time path of income $\{i(t)\}$ to converge to equilibrium level of interest rate (i^*) i.e. $i(t) \to i^*$ as $i_c \to 0$ when $t \to \infty$.

$$i_c \to 0$$

$$\left\{i(0)-\left(\frac{\alpha+\gamma}{\beta+\delta}\right)\right\}e^{\boxed{\left(\frac{\beta+\delta}{\boxed{\rho-\sigma}}\right)}t} \to 0$$

$\left(\frac{\beta+\delta}{\boxed{\rho-\sigma}}\right)$ is the rate of growth/decay of the exponential expression. Therefore, its negativity shall ensure its decay. i.e. $\left(\frac{\beta+\delta}{\boxed{\rho-\sigma}}\right)<0$. It's already known that β and δ are positive and their sum (Numerator; $\beta+\delta$) will also be positive. So, denominator $(\rho-\sigma)$ must be negative. i.e. $(\rho-\sigma)<0$. In comparative terms, $(\rho<\sigma)$. Therefore, the condition for dynamic stability of the time-path of interest rate $i(t)$ is $(\rho<\sigma)$:

$$\left[\underbrace{\left\{\underbrace{i(0)-\overbrace{\boxed{\left(\frac{\alpha+\gamma}{\beta+\delta}\right)}}^{\{\alpha,\beta,\delta,\gamma>0\}}}_{Y(0)\gtreqless Y_p}\right\}}_{\textit{Deviation }(+ve\text{ or }-ve)}\underbrace{e^{\overbrace{\boxed{\left(\frac{\beta+\delta}{\rho-\sigma}\right)}}^{\{\beta,\delta>0\ \&(\rho<\sigma)\}}t}}_{\substack{\textit{Exponential}\\ \textit{Decay}\\ \textit{over time}}}\right]\to 0$$

Therefore,

$$i(t)=\underbrace{\left\{i(0)-\left(\frac{\alpha+\gamma}{\beta+\delta}\right)\right\}e^{\left(\frac{\beta+\delta}{\rho-\sigma}\right)t}}_{\to 0}+\left(\frac{\alpha+\gamma}{\beta+\delta}\right)$$

$$i(t)=\underbrace{\left\{i(0)-\boxed{\left(\frac{\alpha+\gamma}{\beta+\delta}\right)}\right\}e^{\left(\frac{\beta+\delta}{\rho-\sigma}\right)t}}_{\to 0}+\boxed{\left(\frac{\alpha+\gamma}{\beta+\delta}\right)}$$

$$i(t)=\underbrace{\left\{i(0)-\boxed{(i_p)}\right\}e^{\left(\frac{\beta+\delta}{\rho-\sigma}\right)t}}_{\to 0}+\boxed{(i_p)}$$

$$i(t) = \underbrace{\left\{i(0) - \boxed{i_p}\right\} e^{\left(\frac{\beta+\delta}{\rho-\sigma}\right)t}}_{\to 0} + \boxed{i_p}$$

As particular integral is also the equilibrium value of the variable $\boxed{i_p = i^*}$.

$$i(t) = \underbrace{\left\{i(0) - \boxed{i^*}\right\} e^{\left(\frac{\beta+\delta}{\rho-\sigma}\right)t}}_{\to 0} + \boxed{i^*}$$

$$\boxed{i(t) \to \infty}$$

Interpretation: It can be said that provided $\beta > 0, \delta > 0$ with $\rho < \sigma$, regardless of initial interest rate $i(0)$ is greater or smaller than equilibrium interest rate (i^*), the time-path of interest rate shall **converge towards equilibrium** level of interest rate as time is allowed to pass indefinitely. Hence the time-path is **dynamically stable**.

Lesson 23

VARIOUS ECONOMIC DYNAMICS USING DIFFERENTIAL EQUATIONS

TOPIC 094: DYNAMICS OF VALUE OF OIL WELL USING DIFFERENTIAL EQUATIONS

The value of an oil well, which is explored, initially is $V(0) = \$2700$, and is decaying at the rate $\frac{dV}{dt} + 0.04\,V(t) = 28$.

a) What is the general time path that describes the value of this well?

b) What is the future value of the well al time $t = 10$?

Solution: Given the equation for rate of value of oil-well $\frac{dV}{dt} + 0.04\,V(t) = 28$; we can rewrite to compare with the standard form of differential equation:

$$\frac{dV}{dt} + 0.04V = 28$$

$$\underbrace{\frac{dV}{dt}}_{\frac{dy}{dt}} + \underbrace{(0.04)}_{a}\underbrace{V}_{y} = \underbrace{28}_{b}$$

Here $\frac{dy}{dt} = \frac{dV}{dt}$, $y = V$, $a = 0.04$, $b = 28$, $y(0) = V(0) = 2700$.

Choice of Case of the Differential Equation:

As $b = 28$, so $b \neq 0$**:** Non-Homogenous Case.

As $a = 0.04$, so $a \neq 0$**:** Non-Homogenous Case with $a \neq 0$.

Therefore, using the Definite Solution of time-path of $NHC_{a\neq 0}$:

$$y(t) = \left\{y(0) - \frac{b}{a}\right\}e^{-at} + \frac{b}{a}$$

Emphasizing on the values to be substituted:

$$\underline{y}(t) = \left\{\underline{y(0)} - \frac{\boxed{b}}{\boxed{a}}\right\}e^{-\boxed{a}t} + \frac{\boxed{b}}{\boxed{a}}$$

Substituting values of y, a, b and $y(0)$ respectively:

$$\underline{V}(t) = \left\{\underline{V(0)} - \frac{\boxed{28}}{\boxed{0.04}}\right\}e^{-\boxed{0.04}t} + \frac{\boxed{28}}{\boxed{0.04}}$$

$$V(t) = \left(2700 - \frac{28}{0.04}\right)e^{-0.04t} + \frac{28}{0.04}$$

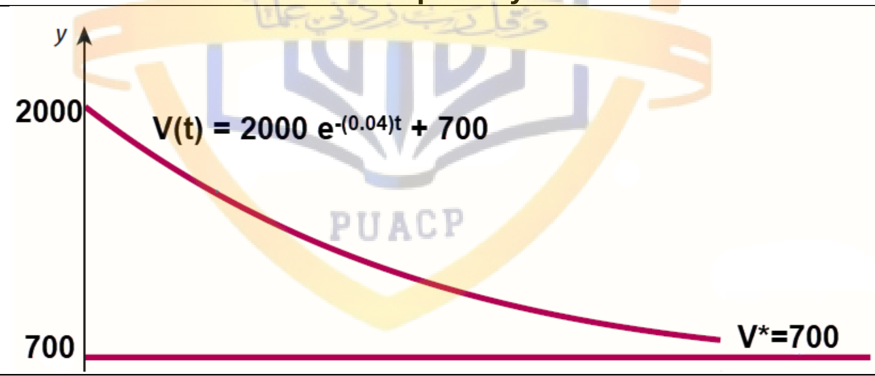
$$V(t) = (2700 - 700)e^{-0.04t} + 700$$

$$\boxed{V(t) = (2000)e^{-0.04t} + 700}$$

It is the definite Time-Path of Value of Oil-Well.

$$V(t) = \underbrace{2000\,e^{-0.04t}}_{V_c} + \underbrace{700}_{V_p}$$

Dynamic Stability: The expression; $V(t) = \boxed{2000\,e^{-0.04t}} + 700$ has an exponential decay term in the box. **i.e.**:

t	Complementary Function (Deviation)	Time-Path
t	$V_c = \boxed{2000\, e^{-0.04t}}$	$V(t) = \boxed{2000\, e^{-0.04t}} + 700$
0	2000	2700
1	1921.58	2621.58
40	403.793	1103.79
80	81.5244	781.524
120	16.4595	716.459
160	3.32311	703.323
200	0.67093	700.671
240	0.13546	700.135
280	0.02735	700.027
320	0.00552	700.006
340	0.00248	700.002
…	…	…
	Remarks	
	Positive deviation decays over time	Time-path **slides and converges** to equilibrium $(V^* = 700)$
	As $t \to \infty$, $V_c \to 0$	And when $V_c \to 0$, $(V_t)\overrightarrow{Converges}(V^*)$
	Graphically	
		
	The straight line shows the equilibrium value of Value of Oil-Well (V^*), while curve shows the deviation from equilibrium (V_c), and this deviation decreases over time and leads the time-path to convergence.	

TOPIC 095: DYNAMICS OF VALUE OF GOLD MINE USING DIFFERENTIAL EQUATIONS

The value of a gold mine is initially $G(0) = \$1,400$ and is decaying at the rate $\frac{dG}{dt} + 0.04\, G(t) = 16$.

a) What is the general time path that describes the value of the gold mine?

b) What is the future value of the gold mine at time $t = 10$?

Solution: Given the rate of change of value of Gold mine $\frac{dG}{dt} + 0.04\,G(t) = 16$, we can rewrite to compare with the standard form of differential equation:

$$\frac{dG}{dt} + 0.04\,G = 16$$

$$\underbrace{\frac{dG}{dt}}_{\frac{dy}{dt}} + \underbrace{(0.04)}_{a}\,\underbrace{G}_{y} = \underbrace{16}_{b}$$

Here $\frac{dy}{dt} = \frac{dG}{dt}$, $y = G$, $a = 0.04$, $b = 16$, $y(0) = G(0) = 1400$.

Choice of Case of the Differential Equation:

As $b = 16$, so $b \neq 0$**:** Non-Homogenous Case.

As $a = 0.04$, so $a \neq 0$**:** Non-Homogenous Case with $a \neq 0$.

Therefore, using the Definite Solution of time-path of $NHC_{a\neq 0}$:

$$y(t) = \left\{y(0) - \frac{b}{a}\right\} e^{-at} + \frac{b}{a}$$

Emphasizing on the values to be substituted:

$$\underline{y}(t) = \left\{\underline{y(0)} - \frac{\boxed{b}}{\boxed{a}}\right\} e^{-\boxed{a}t} + \frac{\boxed{b}}{\boxed{a}}$$

Substituting values of y, a, b and $y(0)$ respectively:

$$\underline{G}(t) = \left\{\underline{G(0)} - \frac{\boxed{16}}{\boxed{0.04}}\right\} e^{-\boxed{0.04}t} + \frac{\boxed{16}}{\boxed{0.04}}$$

$$G(t) = \left(1400 - \frac{16}{0.04}\right) e^{-0.04t} + \frac{16}{0.04}$$

$$G(t) = (1400 - 400)e^{-0.04t} + 400$$

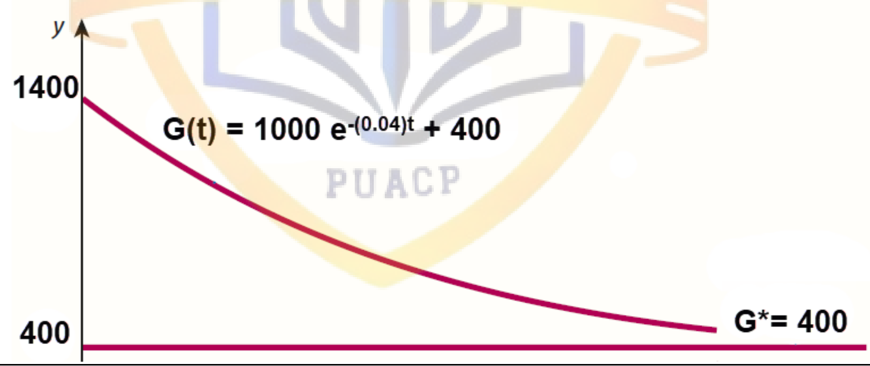
$$\boxed{G(t) = (1000)e^{-0.04t} + 400}$$

It is the definite Time-Path of Value of Gold-Mine.

$$G(t) = \underbrace{1000\,e^{-0.04t}}_{G_c} + \underbrace{400}_{G_p}$$

Dynamic Stability: The expression; $G(t) = \boxed{1000\,e^{-0.04t}} + 400$ has an exponential decay term in the box. **i.e.**:

	Complementary Function (Deviation)	Time-Path
t	$G_c = \boxed{1000\ e^{-0.04t}}$	$G(t) = \boxed{1000\ e^{-0.04t}} + 400$
0	1000	1400
1	960.79	1360.789
40	201.90	601.8965
80	40.762	440.7622
120	8.2297	408.2297
160	1.6616	401.6615
200	0.3355	400.3354
240	0.0677	400.0677
280	0.0137	400.0136
320	0.0028	400.0027
360	0.0006	400.0006
…	…	…

	Remarks	
	Positive deviation decays over time	Time-path **slides and converges** to equilibrium ($G^* = 400$)
	As $t \to \infty$, $G_c \to 0$	And when $G_c \to 0$, $(G_t)\overrightarrow{Converges}(G^*)$
	Graphically	
		
	The straight line shows the equilibrium value of Value of Oil-Well (G^*), while curve shows the deviation from equilibrium (G_c), and this deviation decreases over time and leads the time-path to convergence.	

TOPIC 096: HOTELLING'S RULE USING DIFFERENTIAL EQUATIONS

Hotelling's rule is named after statistician and economic theorist Harold Hotelling.
It studies the net revenue of extraction from a nonrenewable natural resource as a function of time $\pi(t)$.
Hotelling found the optimum point to be where net revenue grows exactly at the rate of interest r.
Thus, the simple rule is:

$$\underbrace{\frac{\pi'(t)}{\pi(t)}}_{\substack{Net\\ revenue}} = \underbrace{r}_{\substack{rate\ of\\ Interest}}$$

Rewriting the Hotelling's Rule:

$$\frac{\pi'(t)}{\pi(t)} = r$$

$$\pi'(t) = r \times \pi(t)$$

$$\frac{d\{\pi(t)\}}{dt} = r \times \pi(t)$$

In simple terms:

$$\frac{d\pi}{dt} = r\pi$$

Rewriting to compare with standard form:

$$\frac{d\pi}{dt} - r\pi = 0$$

$$\underbrace{\frac{d\pi}{dt}}_{\frac{dy}{dt}} - \underbrace{r}_{a}\underbrace{\pi}_{y} = \underbrace{0}_{b}$$

Here $\frac{dy}{dt} = \frac{d\pi}{dt}, y = \pi, a = r, b = 0$

Choice of Case of the Differential Equation:

$b = 0$ **– Homogenous Case.**

$a = r$; interest rate is usually a positive number $r > 0$ i.e. $a \neq 0$ **– Homogenous Case with** $a \neq 0$.

Therefore, using the definite Solution of time-path of HC:

$$y(t) = \{y(0)\}e^{-at}$$

Emphasizing on the values to be substituted:

$$\underline{y}(t) = \left\{\underline{y(0)}\right\} e^{-\boxed{a}t}$$

Substituting values of y and a, respectively:

$$\underline{\pi}(t) = \left\{\underline{\pi(0)}\right\} e^{-\boxed{r}t}$$

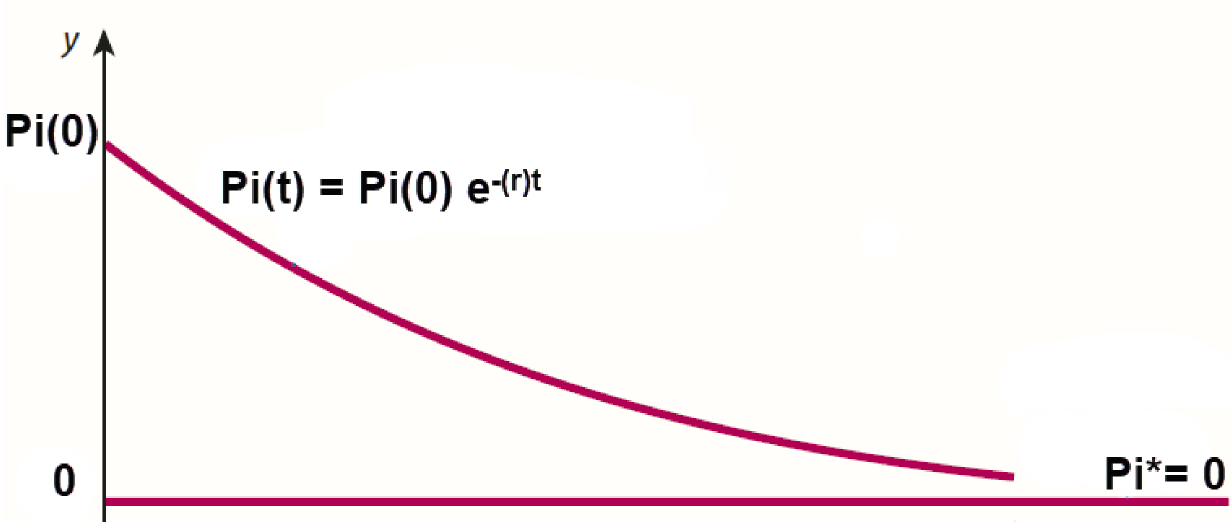
$$\boxed{\pi(t) = \{\pi(0)\}e^{-rt}}$$

Definite Time-path for net revenue in Dynamic Sense.

Dynamic Stability: For the dynamic stability of the time-path of First Order Differential Equation, the deviation expression should fade. In homogenous case (HC), the particular solution is zero; $\pi_p = 0$. Therefore, the convergence should take place towards zero (0). In this case, we have a negative exponent $\pi(t) = \{\pi(0)\}e^{\boxed{-r}t}$which shall approach to zero (0), when $t \to \infty$. In a differential equation, deviation component is measured by (π_c). Therefore, for dynamic stability; $\pi_c \to 0$ as $t \to \infty$.

$$\left[\underbrace{\{\pi(0)\}}_{Deviation} \quad \underbrace{e^{\overbrace{\boxed{-(r)}}^{(r>0)}t}}_{\substack{Exponential\\ Decay\\ over\ time}}\right] \to 0$$

Therefore, the time-path of net revenue shall **converge towards equilibrium** as time is allowed to pass indefinitely. Hence the time-path is **dynamically stable**.



TOPIC 097: FUTURE VALUE OF INVESTMENT USING DIFFERENTIAL EQUATIONS

An initial amount $\boldsymbol{D(0)} = \$\boldsymbol{1,300}$ deposited in a bank is growing according to the equation:

$$\frac{dD}{dt} - 0.05\,D(t) = 15$$

a) What is the general time path that describes the growth of this account?
b) What is the future value of this account at time $\boldsymbol{t = 6}$?

Solution: Given the rate of change of value of deposit $\frac{dD}{dt} - 0.05\,D\boxed{(t)} = 15$, we can rewrite to compare with the standard form of differential equation:

$$\frac{dD}{dt} + 0.05\,D = 15$$

$$\underbrace{\frac{dD}{dt}}_{\frac{dy}{dt}} + \underbrace{(0.05)}_{a}\,\underbrace{D}_{y} = \underbrace{15}_{b}$$

Here $\frac{dy}{dt} = \frac{dD}{dt}$, $y = D$, $a = 0.05$, $b = 15$, $y(0) = D(0) = 1300$.

Choice of Case of the Differential Equation:

As $b = 15$, so $b \neq 0$: Non-Homogenous Case.

As $a = 0.05$, so $a \neq 0$: Non-Homogenous Case with $a \neq 0$.

Therefore, using the Definite Solution of time-path of $NHC_{a\neq 0}$:

$$y(t) = \left\{y(0) - \frac{b}{a}\right\} e^{-at} + \frac{b}{a}$$

Emphasizing on the values to be substituted:

$$\underline{y}(t) = \left\{\underline{y(0)} - \frac{\boxed{b}}{\boxed{a}}\right\} e^{-\boxed{a}t} + \frac{\boxed{b}}{\boxed{a}}$$

Substituting values of y, a, b and $y(0)$ respectively:

$$\underline{D}(t) = \left\{\underline{D(0)} - \frac{\boxed{15}}{\boxed{0.05}}\right\} e^{-\boxed{0.05}t} + \frac{\boxed{15}}{\boxed{0.05}}$$

$$D(t) = (1300 - 300)e^{-0.05t} + 300$$

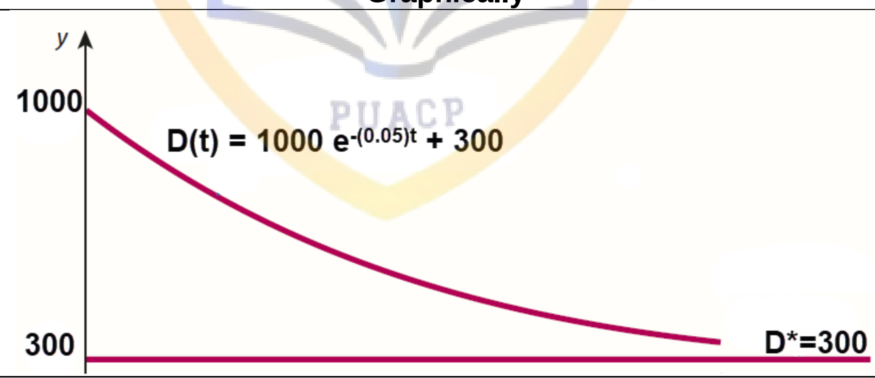
$$D(t) = (1300 - 300)e^{-0.05t} + 300$$

$$\boxed{D(t) = (1000)e^{-0.05t} + 300}$$

It is the definite Time-Path of Value of Deposit.

$$D(t) = \underbrace{1000\, e^{-0.05t}}_{D_c} + \underbrace{300}_{D_p}$$

Dynamic Stability: The expression; $D(t) = \boxed{1000\, e^{-0.05t}} + 300$ has an exponential decay term in the box. **i.e.**:

	Complementary Function (Deviation)	**Time-Path**
t	$D_c = \boxed{1000\, e^{-0.05t}}$	$D(t) = \boxed{1000\, e^{-0.05t}} + 300$
0	1000	1300
1	951.2	1251.22
30	223.1	523.130
60	49.78	349.787
90	11.10	311.108
120	2.478	302.478
150	0.553	300.553
180	0.123	300.123
210	0.027	300.027
240	0.006	300.006
270	0.001	300.001
…	…	…
	Remarks	
	Positive deviation decays over time	Time-path **slides and converges** to equilibrium $(D^* = 400)$
	As $t \to \infty$, $D_c \to 0$	And when $D_c \to 0$, $(D_t)\overrightarrow{Converges}(D^*)$
	Graphically	
		
	The straight line shows the equilibrium value of Value of Deposit (D^*), while curve shows the deviation from equilibrium (D_c), and this deviation decreases over time and leads the time-path to convergence.	

<u>TOPIC 098: DYNAMICS OF MONTHLY SALES USING DIFFERENTIAL EQUATIONS</u>

A model predicts that monthly sales, $S(t)$, of a new product after t months satisfy the differential equation:

$\frac{dS}{dt} = k(M - S)$ with initial condition $S(0) = C$.

Where k, M and C are positive constants.

Solution: Given the rate of change of monthly sales; $\frac{dS}{dt} = k(M - S)$, we can rewrite it for the sake of comparison with first order differential equation.

$$\frac{dS}{dt} = k(M - S)$$
$$\frac{dS}{dt} = kM - kS$$
$$\underbrace{\frac{dS}{dt}}_{\frac{dy}{dt}} + \underbrace{k}_{a}\underbrace{S}_{y} = \underbrace{kM}_{b}$$

Here $\frac{dy}{dt} = \frac{dS}{dt}, y = S, a = k, b = kM, y(0) = S(0) = C$

Choice of Case of the Differential Equation:

$b = kM$; As $k > 0$ and $M > 0$ (both positive constants), their product $(k \times M)$ will be non-zero positive constant. i.e. $b \neq 0$ **– Non-Homogenous Case**.

$a = k$; $k > 0$ (non-zero positive constants). i.e. $a \neq 0$ **– Non-Homogenous Case with $a \neq 0$.**

Therefore, using the definite Solution of time-path of $NHC_{a\neq 0}$:

$$y(t) = \left\{y(0) - \frac{b}{a}\right\}e^{-at} + \frac{b}{a}$$

Emphasizing on the values to be substituted:

$$\underline{y}(t) = \left\{\underline{y}(0) - \frac{\boxed{b}}{\boxed{a}}\right\}e^{-\boxed{a}t} + \frac{\boxed{b}}{\boxed{a}}$$

Substituting values of y, a and b, respectively:

$$\underline{S}(t) = \left\{\underline{S}(0) - \frac{\boxed{kM}}{\boxed{k}}\right\}e^{-\boxed{k}t} + \frac{\boxed{kM}}{\boxed{k}}$$
$$S(t) = \left\{\boxed{S(0)} - \frac{\cancel{k}M}{\cancel{k}}\right\}e^{-kt} + \frac{\cancel{k}M}{\cancel{k}}$$
$$S(t) = (\boxed{C} - M)e^{-kt} + M$$
$$\boxed{S(t) = (C - M)e^{-kt} + M}$$

Definite Time-path for Monthly Sales in Dynamic Sense.

Dynamic Stability: For the dynamic stability of the time-path of First Order Differential Equation, it should converge towards equilibrium. We know that equilibrium is particular solution (S_p). Whereas convergence happens when deviation from equilibrium fades to zero. In a differential equation, deviation component is measured by (S_c). Therefore, for dynamic stability; $S_c \to 0$ as $t \to \infty$.

Using the Definite Time-path for Income level in Dynamic Sense.

$$S(t) = \underbrace{(C - M)e^{-kt}}_{S_c} + \underbrace{M}_{S_p}$$

Need for time path of Monthly Sales $\{S(t)\}$ to converge to equilibrium income (S^*) i.e. $S(t) \to S^*$ as $S_c \to 0$ when $t \to \infty$.

$$S_c \to 0$$
$$(C - M)e^{-kt} \to 0$$
$$\left[\underbrace{\left\{\underbrace{C - \overbrace{\boxed{M}}^{\{M>0\}}}_{\{S(0)=C\}\gtrless S_p}\right\}}_{\substack{Deviation\\(+ve\ or -ve)}} \underbrace{e^{-\overbrace{\boxed{k}}^{\{k>0\}}t}}_{\substack{Exponential\\Decay\\over\ time}}\right] \to 0$$

Therefore,

$$S(t) = \underbrace{(C - M)e^{-kt}}_{\to 0} + M$$

$$S(t) = \underbrace{\left(C - \boxed{M}\right)e^{-kt}}_{\to 0} + \boxed{M}$$

$$S(t) = \underbrace{\left(C - \boxed{S_p}\right)e^{-kt}}_{\to 0} + \boxed{S_p}$$

As particular integral is also the equilibrium value of the variable $\boxed{S_p = S^*}$.

$$S(t) = \underbrace{\left(C - \boxed{S^*}\right)e^{-kt}}_{\to 0} + \boxed{S^*}$$

$$\boxed{S(t) \to S^*}$$

Interpretation: It can be said that provided $C > 0, k > 0$, regardless of initial national income $S(0)$ is greater or smaller than equilibrium monthly sales (S^*), the time-path of monthly sales shall **converge towards equilibrium** level of monthly sales as time is allowed to pass indefinitely. Hence the time-path is **dynamically stable**.

TOPIC 099: DYNAMICS OF POPULATION USING DIFFERENTIAL EQUATIONS

The population, N, of a small country is currently 5 million. It is expected that this population will grow according to the differential equation.

$$\frac{dN}{dt} = 0.015N$$

where t is measured in years.

Solution: Current population of the country is 5 million. It implies; $N(0) = 5,000,000$.

Given the rate of change of population; $\frac{dN}{dt} = 0.015N$, rewrite to compare with standard form:

$$\frac{dN}{dt} = 0.015N$$

$$\frac{dN}{dt} - 0.015N = 0$$

$$\frac{dN}{dt} + (-0.015)N = 0$$

$$\underbrace{\frac{dN}{dt}}_{\frac{dy}{dt}} + \underbrace{(-0.015)}_{a}\underbrace{N}_{y} = \underbrace{0}_{b}$$

Here $\frac{dy}{dt} = \frac{dN}{dt}, y = N, a = -0.015, b = 0, y(0) = N(0) = 5000000$

Choice of Case of the Differential Equation:

$b = 0$; **– Homogenous Case**.

$a = -0.015$; i.e. $a \neq 0$ **– Homogenous Case with $a \neq 0$**.

Therefore, using the definite Solution of time-path of HC:

$$y(t) = \{y(0)\}e^{-at}$$

Emphasizing on the values to be substituted:

$$\underline{y}(t) = \left\{\underline{y(0)}\right\}e^{-\boxed{a}t}$$

Substituting values of y and a, respectively:

$$\underline{N}(t) = \left\{\underline{N(0)}\right\}e^{\boxed{-}\boxed{(\boxed{-}0.015)}t}$$

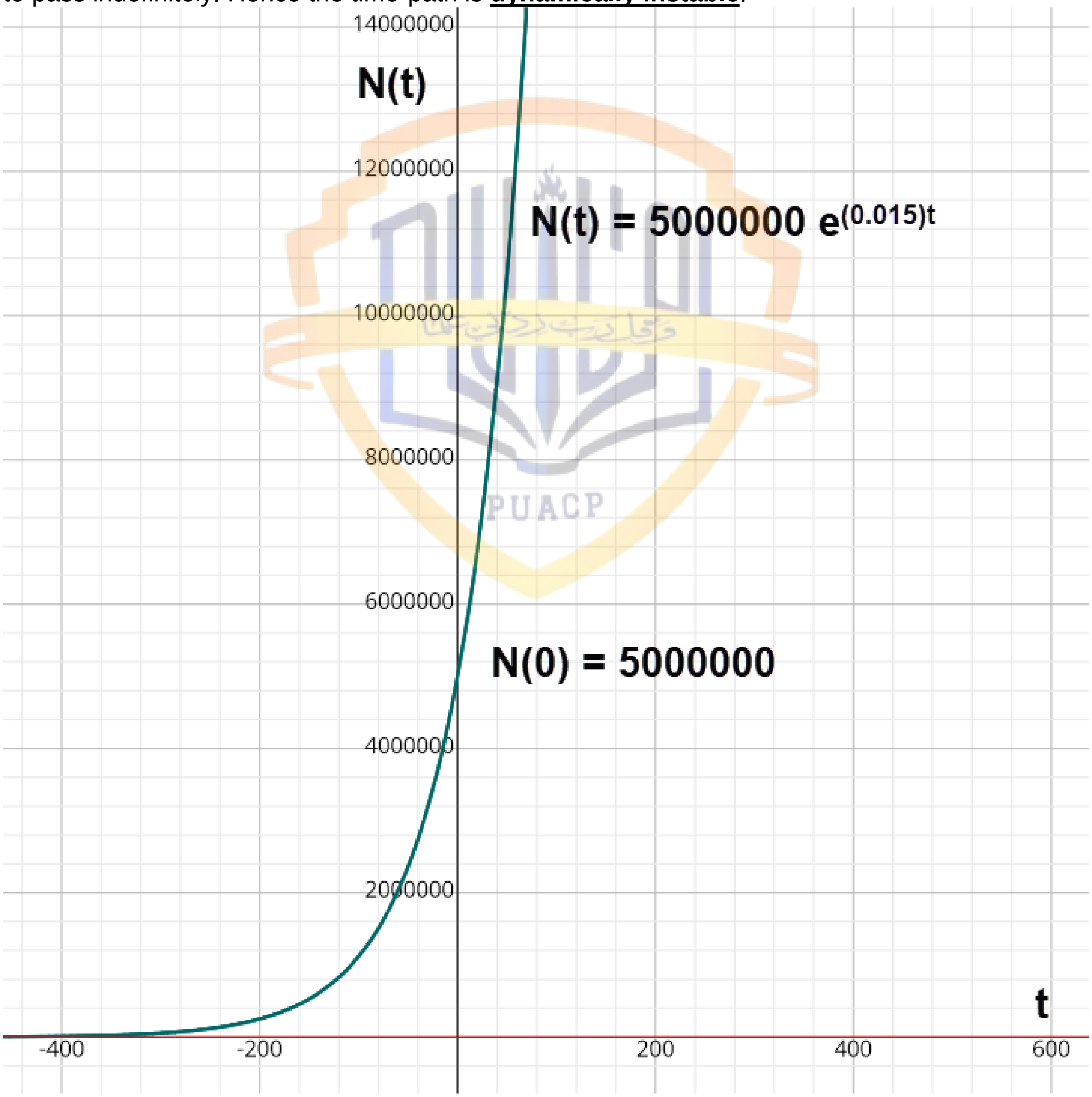
$$\boxed{N(t) = (5000000)e^{0.015t}}$$

Definite Time-path for net revenue in Dynamic Sense.

Dynamic Stability: For the dynamic stability of the time-path of First Order Differential Equation, the deviation expression should fade. In homogenous case $(\boldsymbol{HC})$, the particular solution is zero; $\boldsymbol{N_p} = \mathbf{0}$. Therefore, the convergence should take place towards zero $(\mathbf{0})$. In this case, we have a positive exponent $\boldsymbol{N}(\boldsymbol{t}) = (\mathbf{5000000})\boldsymbol{e}^{\boxed{\mathbf{0.015}}\boldsymbol{t}}$which shall approach to infinity (∞), when $\boldsymbol{t} \to \infty$. In a differential equation, deviation component is measured by $(\boldsymbol{N_c})$. Here, there is dynamic instability; $\boldsymbol{N_c} \to \infty$ as $\boldsymbol{t} \to \infty$.

$$\left[\underbrace{\{\mathbf{5000000}\}}_{\textit{Deviation}}\ \underbrace{\boldsymbol{e}^{\overbrace{\boxed{(\mathbf{0.015})}}^{(r>0)}t}}_{\substack{\textit{Exponential}\\ \textit{Growth}\\ \textit{over time}}}\right] \to \mathbf{0}$$

Therefore, the time-path of population shall **<u>diverge away from equilibrium</u>** as time is allowed to pass indefinitely. Hence the time-path is **<u>dynamically instable</u>**.



Policy Problem: From policy perspective, its important to know about the time the population will take to get double. i.e. $\{\boldsymbol{N(T) = 10,000,000}\}_{\boldsymbol{T=?}}$
Using the solution $\boldsymbol{N(t)} = (\boldsymbol{5000000})\boldsymbol{e^{0.015t}}$ for substitution, we get:

$$\boldsymbol{N(T)} = (\boldsymbol{5000000})\boldsymbol{e^{0.015\boxed{(T)}}}$$

$$\boldsymbol{10000000} = (\boldsymbol{5000000})\boldsymbol{e^{0.015\boxed{T}}}$$

$$\frac{\boldsymbol{10000000}}{\boldsymbol{5000000}} = \boldsymbol{e^{0.015\boxed{T}}}$$

$$\boldsymbol{e^{0.015\boxed{T}}} = \boldsymbol{2}$$

Raising both sides to exponent:

$$\cancel{\boldsymbol{ln}}\left(\boldsymbol{e^{0.015\boxed{T}}}\right) = \boldsymbol{ln(2)}$$

$$\boldsymbol{0.015\boxed{T} = 0.6931471805599}$$

$$\boxed{\boldsymbol{T}} = \frac{\boldsymbol{0.6931471805599}}{\boldsymbol{0.015}}$$

$$\boxed{\boldsymbol{T = 46.2}}$$

Interpretation: It will take $\boldsymbol{46.2\,years}$ approximately for the population of the country to get double i.e. from $\boldsymbol{5,000,000}$ to $\boldsymbol{10,000,000}$.

Lesson 24

VARIOUS ECONOMIC DYNAMICS USING DIFFERENTIAL EQUATIONS (CONTINUED)

TOPIC 100: DYNAMICS OF GOVERNMENT SPENDING USING DIFFERENTIAL EQUATIONS

Government expenditure, $\boldsymbol{G}$, satisfies the differential equation:

$$\frac{dG}{dt} = -0.05G$$

Where, $\boldsymbol{G}$ is measured in billions of dollars and time $\boldsymbol{t}$ is in years. The initial condition is $\boldsymbol{G(0) = 500}$.

(a) Solve this equation to express $\boldsymbol{G}$ in terms of $\boldsymbol{t}$.

(b) Find the value of $\boldsymbol{G}$ when $\boldsymbol{t = 6}$. Round your answers to three significant figures.

Solution: Given the rate of change of Government Expenditure; $\frac{dG}{dt} = -0.05G$, rewrite to compare with standard form:

$$\frac{dG}{dt} = -0.05G$$

$$\frac{dG}{dt} + 0.05G = 0$$

$$\frac{dG}{dt} + 0.05G = 0$$

$$\underbrace{\frac{dG}{dt}}_{\frac{dy}{dt}} + \underbrace{(0.05)}_{a}\underbrace{G}_{y} = \underbrace{0}_{b}$$

Here $\frac{dy}{dt} = \frac{dG}{dt}, y = G, a = 0.05, b = 0, y(0) = G(0) = 500$

Choice of Case of the Differential Equation:

$\boldsymbol{b = 0}$; **– Homogenous Case**.

$\boldsymbol{a = 0.05}$; i.e. $\boldsymbol{a \neq 0}$ **– Homogenous Case with $\boldsymbol{a \neq 0}$**.

Therefore, using the definite Solution of time-path of $\boldsymbol{HC}$:

$$y(t) = \{y(0)\}e^{-at}$$

Emphasizing on the values to be substituted:

$$\underline{y}(t) = \left\{\underline{y(0)}\right\}e^{-\boxed{a}t}$$

Substituting values of $\boldsymbol{y}$ and $\boldsymbol{a}$, respectively:

$$\underline{G}(t) = \left\{\underline{G(0)}\right\}e^{-\boxed{(0.05)}t}$$

$$G(t) = \left(\underline{500}\right)e^{-0.05t}$$

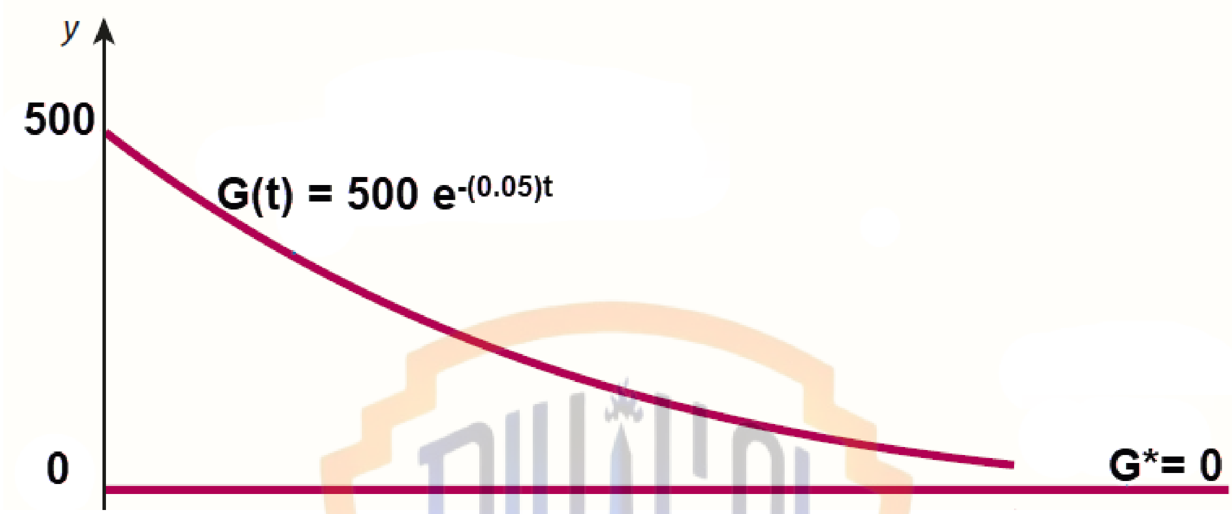
$$\boxed{G(t) = (500)e^{-0.05t}}$$

Definite Time-path for Government Expenditure in Dynamic Sense.

Dynamic Stability: For the dynamic stability of the time-path of First Order Differential Equation, the deviation expression should fade. In homogenous case ($\boldsymbol{HC}$), the particular solution is zero; $\boldsymbol{G_p = 0}$. Therefore, the convergence should take place towards zero ($\boldsymbol{0}$). In this case, we have a negative exponent $\boldsymbol{G(t) = (500)e^{-\boxed{0.05}t}}$which shall approach to zero ($\boldsymbol{0}$), when $\boldsymbol{t \to \infty}$. In a differential equation, deviation component is measured by ($\boldsymbol{G_c}$). Therefore, for dynamic stability; $\boldsymbol{G_c \to 0}$ as $\boldsymbol{t \to \infty}$.

$$\left[\underbrace{\{500\}}_{Deviation}\ \underbrace{e^{\boxed{-(0.05)}t}}_{\substack{Exponential\\ Decay\\ over\ time}}\right] \rightarrow 0$$

Therefore, the time-path of Government Expenditure shall **converge towards equilibrium** as time is allowed to pass indefinitely. Hence the time-path is **dynamically stable**.



Policy Problem: From policy perspective, it's important to know about the level of government expenditure after some time.
For example; after **6** years: $\{G(6) = ?\}$
Using the solution $G(t) = (500)e^{-0.05t}$ for substitution, we get:

$$G(6) = (500)e^{-0.05\boxed{(6)}}$$
$$G(6) = (500)e^{-0.3}$$
$$\boxed{G(6) = \$\,370.409\ billion}$$

TOPIC 101: DYNAMICS OF FUEL CONSUMPTION USING DIFFERENTIAL EQUATIONS

Let $F(x)$ denote the number of litre(s) of fuel left in an aircraft's fuel tank if it has flown x km. Suppose that $F(x)$ satisfies the following differential equation: $F'(x) = -aF(x) - b$. Here, the fuel consumption per km is a constant $b > 0$. The term $-aF(x)$, with $a > 0$, is due to the weight of the fuel.
Numerically speaking, the differential equation is: $F'(x) = -0.001F(x) - 8.$ The initial level of fuel in the plane is $F(0) = 12000$:
Find the solution of the equation with $F(0) = F_0$.
Solution: Rate of change of remaining fuel; $F'(x) = -0.001F(x) - 8$.
Rewriting equation to compare with the standard form:

$$F'(x) = -0.001F(x) - 8$$
$$F'(x) + 0.001F(x) = -8$$
$$\underbrace{\frac{dF}{dx}}_{\frac{dy}{dx}} + \underbrace{(0.001)}_{a}\underbrace{F}_{y} = \underbrace{-8}_{b}$$

Here $\frac{dy}{dt} = \frac{dF}{dt}$, $y = F$, $a = 0.001$, $b = -8$, $y(0) = F(0) = 12000$.

Choice of Case of the Differential Equation:

As $\boldsymbol{b}=-\mathbf{8}$, so $\boldsymbol{b}\neq\mathbf{0}$**:** Non-Homogenous Case.

As $\boldsymbol{a}=\mathbf{0.001}$, so $\boldsymbol{a}\neq\mathbf{0}$**:** Non-Homogenous Case with $\boldsymbol{a}\neq\mathbf{0}$.

Therefore, using the Definite Solution of time-path of $\boldsymbol{NHC_{a\neq 0}}$:

$$y(t)=\left\{y(0)-\frac{b}{a}\right\}e^{-at}+\frac{b}{a}$$

Emphasizing on the values to be substituted:

$$\underline{y}(t)=\left\{\underline{y(0)}-\frac{\boxed{b}}{\boxed{a}}\right\}e^{-\boxed{a}t}+\frac{\boxed{b}}{\boxed{a}}$$

Substituting values of $\boldsymbol{y}$, $\boldsymbol{a}$**,** $\boldsymbol{b}$ and $\boldsymbol{y(0)}$ respectively:

$$\underline{F}(t)=\left\{\underline{F(0)}-\frac{\boxed{(-8)}}{\boxed{0.001}}\right\}e^{-\boxed{0.001}t}+\frac{\boxed{(-8)}}{\boxed{0.001}}$$

$$F(t)=\left\{\underline{12000}\boxed{-}\frac{(\boxed{-}8)}{0.001}\right\}e^{-(0.001)t}\boxed{+}\frac{(\boxed{-}8)}{0.001}$$

$$F(t)=\left\{12000\boxed{+}\frac{8}{0.001}\right\}e^{-(0.001)t}\boxed{-}\frac{8}{0.001}$$

$$F(t)=\{12000+8000\}e^{-(0.001)t}-8000$$

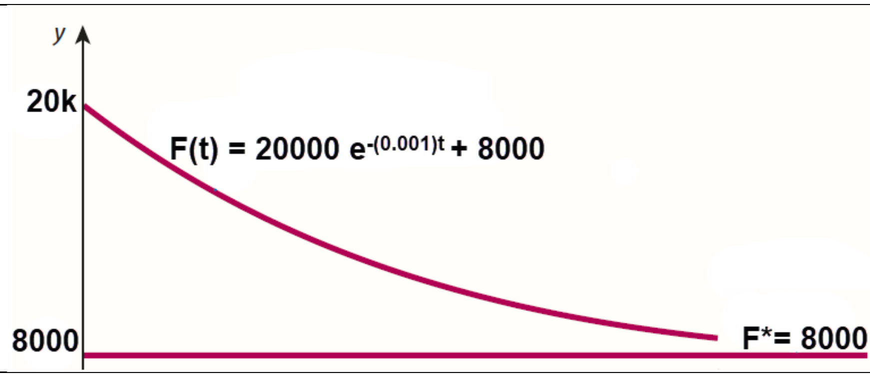
$$\boxed{F(t)=(20000)e^{-(0.001)t}-8000}$$

It is the definite Time-Path of remaining Fuel.

$$F(t)=\underbrace{20000\,e^{-0.001t}}_{F_c}-\underbrace{8000}_{F_p}$$

Dynamic Stability: The expression; $F(t)=\boxed{20000\,e^{-0.001t}}+8000$ has an exponential decay term in the box. **i.e.**:

t	Complementary Function (Deviation) $F_c=\boxed{20000\,e^{-0.001t}}$	Time-Path $F(t)=\boxed{20000\,e^{-0.001t}}+8000$
0	20000	28000
1	19980.0	27980.0
2	19960.0	27960.0
3	19940.0	27940.0
4	19920.1	27920.1
5	19900.2	27900.2
6	19880.3	27880.3
7	19860.4	27860.4
8	19840.6	27840.6
9	19820.8	27820.8
10	19800.9	27800.9
…	…	…
	Remarks	
	Positive deviation decays over time	Time-path **slides and converges** to equilibrium ($F^*=8000$)
	As $t\to\infty$, $F_c\to 0$	And when $F_c\to 0$, $(F_t)\overrightarrow{Converges}(F^*)$
	Graphically	



The straight line shows the equilibrium value of remaining Fuel (F^*), while curve shows the deviation from equilibrium (F_c), and this deviation decreases over time and leads the time-path to convergence.

TOPIC 102: DYNAMICS OF UNEMPLOYMENT USING DIFFERENTIAL EQUATIONS

If unemployment data is measured in thousands, unemployment in a small economy is measured by following differential equation:

$$\frac{dU}{dt} = -3U + 180$$

Here U is the unemployment level (in thousands) and t is time in years. If unemployment is 150 (in thousands) at the beginning of analysis, find the time-path of unemployment in the economy.
Solution: Given the rate of unemployment in the economy;

$$\frac{dU}{dt} = -3U + 180$$

we can rewrite compare it with the standard form of differential equation.

$$\frac{dU}{dt} + 3U = 180$$

$$\underbrace{\frac{dU}{dt}}_{\frac{dy}{dt}} + \underbrace{3}_{a}\underbrace{U}_{y} = \underbrace{180}_{b}$$

Here $\frac{dy}{dt} = \frac{dU}{dt}$, $y = U$, $a = 3$, $b = 180$, $y(0) = U(0) = 150$.

Choice of Case of the Differential Equation:

As $b = 180$, so $b \neq 0$**:** Non-Homogenous Case.

As $a = 3$, so $a \neq 0$**:** Non-Homogenous Case with $a \neq 0$.

Therefore, using the Definite Solution of time-path of $NHC_{a\neq0}$:

$$y(t) = \left\{y(0) - \frac{b}{a}\right\}e^{-at} + \frac{b}{a}$$

Emphasizing on the values to be substituted:

$$\underline{y}(t) = \left\{\underline{y(0)} - \frac{\boxed{b}}{\boxed{a}}\right\}e^{-\boxed{a}t} + \frac{\boxed{b}}{\boxed{a}}$$

Substituting values of y, a, b and $y(0)$ respectively:

$$\underline{U}(t) = \left\{\underline{U(0)} - \frac{\boxed{180}}{\boxed{3}}\right\}e^{-\boxed{3}t} + \frac{\boxed{180}}{\boxed{3}}$$

$$U(t) = \left\{\underline{150} - \frac{\boxed{180}}{\boxed{3}}\right\}e^{-\boxed{3}t} + \frac{\boxed{180}}{\boxed{3}}$$

$$U(t) = (150 - 60)e^{-3t} + 60$$

$$\boldsymbol{U(t) = (90)e^{-3t} + 60}$$

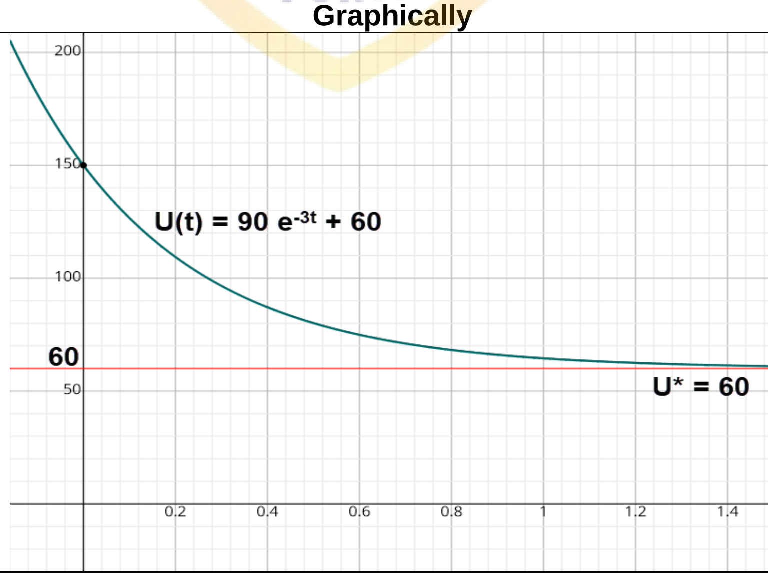
$$\boxed{\boldsymbol{U(t) = (90)e^{-3t} + 60}}$$

It is the definite Time-Path of Unemployment.

$$\boldsymbol{U(t) = \underbrace{90\ e^{-3t}}_{U_c} + \underbrace{60}_{U_p}}$$

Dynamic Stability: The expression; $\boldsymbol{U(t)} = \boxed{\boldsymbol{90\ e^{-3t}}} + \boldsymbol{60}$ has an exponential decay term in the box. **i.e.**:

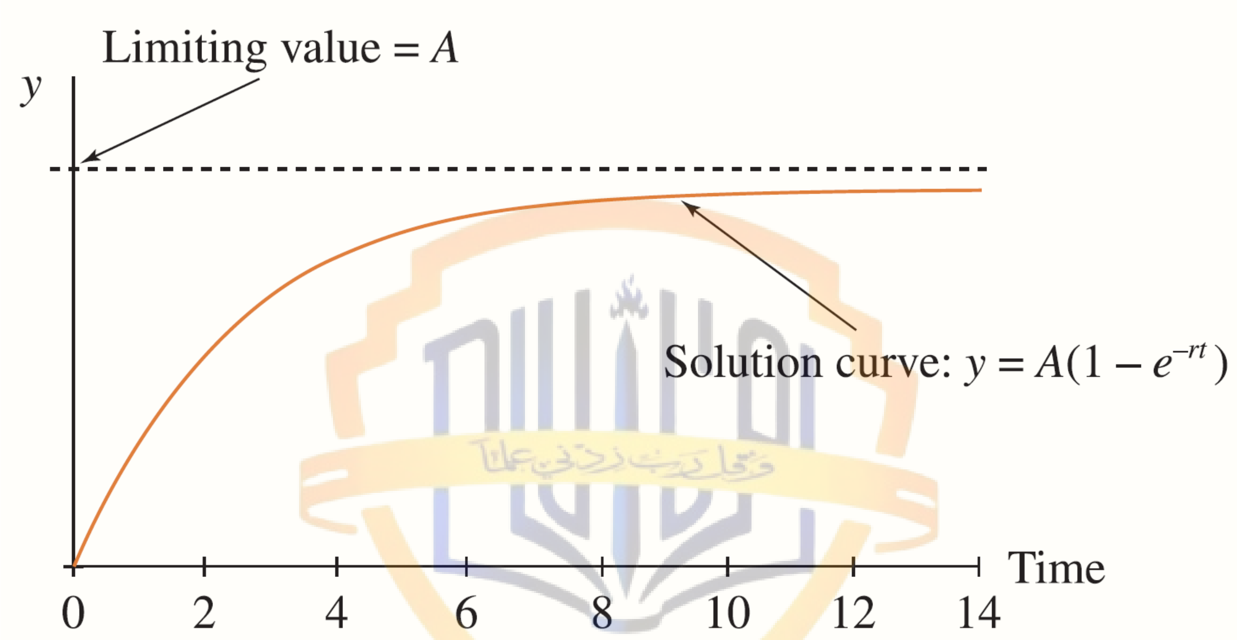
	Complementary Function (Deviation)	**Time-Path**
$\boldsymbol{t}$	$\boldsymbol{U_c} = \boxed{\boldsymbol{90\ e^{-3t}}}$	$\boldsymbol{U(t)} = \boxed{\boldsymbol{90\ e^{-3t}}} + \boldsymbol{60}$
0	90	150
1	4.480836153108	64.480836153108
2	0.223087695900	60.223087695900
3	0.011106882368	60.011106882368
4	0.000552979112	60.000552979112
5	0.000027531209	60.000027531209
6	0.000001370698	60.000001370698
7	0.000000068243	60.000000068243
8	0.000000003398	60.000000003398
9	0.000000000169	60.000000000169
10	0.000000000008	60.000000000008
…	…	…

	Remarks	
	Positive deviation decays over time	Time-path **slides and converges** to equilibrium ($\boldsymbol{U^* = 60}$)
	As $\boldsymbol{t} \to \infty$, $\boldsymbol{U_c} \to \boldsymbol{0}$	And when $\boldsymbol{U_c} \to \boldsymbol{0}$, $(\boldsymbol{U_t})\overrightarrow{Converges}(\boldsymbol{U^*})$
	Graphically	
		

	The straight line shows the equilibrium value of Unemployment ($\boldsymbol{U^*}$), while curve shows the deviation from equilibrium ($\boldsymbol{U_c}$), and this deviation decreases over time and leads the time-path to convergence.

TOPIC 103: LIMITED GROWTH USING DIFFERENTIAL EQUATIONS

Situations where growth of a variable is fast initially but levels-off (stabilize) just short of a maximum limit. Examples include mobile adoption, wireless internet etc.
Using differential equations of the type $\frac{dy}{dt} = \boldsymbol{r(A-y)}$ represent a similar situation where $\boldsymbol{A}$ is the maximum limit. Graphically it is:



The rate at which the volume of sales ($\boldsymbol{Q}$) for a new type of printer increases after an advertising campaign is given by the equation $\frac{dQ}{dt} = \boldsymbol{0.04(700 - Q)}$, given that $\boldsymbol{Q = 0}$ at $\boldsymbol{t = 0}$. $\boldsymbol{Q}$ is the number of printers sold, $\boldsymbol{t}$ is time in years.
(a) Solve the differential equation to obtain an expression for $\boldsymbol{Q}$ in terms of $\boldsymbol{t}$.
(b) Sketch the solution curve, hence describe verbally how the volume of sales should increase in time.
Solution: Given the rate of change of Volume of Sales; $\frac{dQ}{dt} = \boldsymbol{0.04(700 - Q)}$, we can rewrite it to compare with standard form of differential equation.

$$\frac{dQ}{dt} = \boldsymbol{0.04(700 - Q)}$$
$$\frac{dQ}{dt} = \boldsymbol{28 - 0.04Q}$$
$$\underbrace{\frac{dQ}{dt}}_{\frac{dy}{dt}} + \underbrace{\boldsymbol{0.04}}_{a}\underbrace{\boldsymbol{Q}}_{y} = \underbrace{\boldsymbol{28}}_{b}$$

Here $\frac{dy}{dt} = \frac{dQ}{dt}$, $\boldsymbol{y = Q}$, $\boldsymbol{a = 0.04}$, $\boldsymbol{b = 28}$, $\boldsymbol{y(0) = Q(0) = 0}$.

Choice of Case of the Differential Equation:
As $\boldsymbol{b = 28}$, so $\boldsymbol{b \neq 0}$: Non-Homogenous Case.

As $\boldsymbol{a} = \mathbf{0.04}$, so $\boldsymbol{a} \neq \mathbf{0}$**:** Non-Homogenous Case with $\boldsymbol{a} \neq \mathbf{0}$.
Therefore, using the Definite Solution of time-path of $\boldsymbol{NHC_{a\neq 0}}$:

$$\boldsymbol{y(t) = \left\{y(0) - \frac{b}{a}\right\} e^{-at} + \frac{b}{a}}$$

Emphasizing on the values to be substituted:

$$\boldsymbol{\underline{y}(t) = \left\{\underline{y(0)} - \frac{\boxed{b}}{\boxed{a}}\right\} e^{-\boxed{a}t} + \frac{\boxed{b}}{\boxed{a}}}$$

Substituting values of $\boldsymbol{y}$, $\boldsymbol{a}$**,** $\boldsymbol{b}$ and $\boldsymbol{y(0)}$ respectively:

$$\boldsymbol{\underline{Q}(t) = \left\{\underline{Q(0)} - \frac{\boxed{28}}{\boxed{0.04}}\right\} e^{-\boxed{0.04}t} + \frac{\boxed{28}}{\boxed{0.04}}}$$

$$\boldsymbol{Q(t) = \left\{\underline{0} - \frac{28}{0.04}\right\} e^{-0.04t} + \frac{28}{0.04}}$$

$$\boldsymbol{Q(t) = (0 - 700)e^{-0.04t} + 700}$$

$$\boxed{\boldsymbol{Q(t) = (-700)e^{-0.04t} + 700}}$$

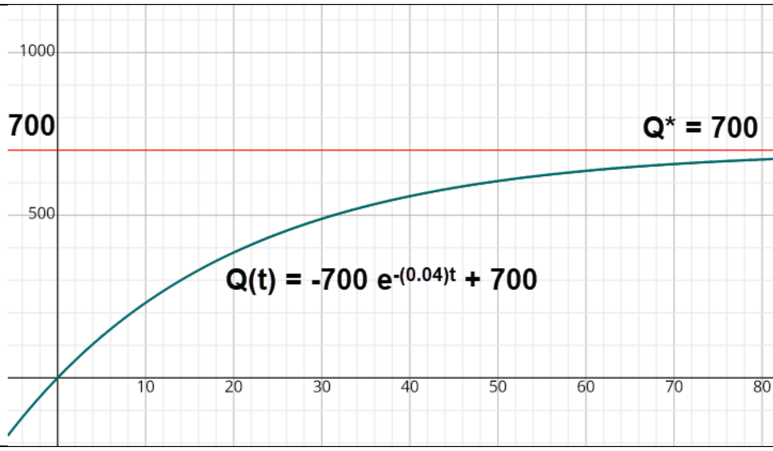
It is the definite Time-Path of Quantity Supplied.

$$\boldsymbol{Q(t) = \underbrace{-700\, e^{-0.04t}}_{Q_c} + \underbrace{700}_{Q_p}}$$

Dynamic Stability: The expression; $\boldsymbol{Q(t)} = \boxed{\boldsymbol{-700\, e^{-0.04t}}} + \mathbf{700}$ has an exponential decay term in the box. **i.e.**:

$\boldsymbol{t}$	**Complementary Function (Deviation)** $\boldsymbol{Q_c} = \boxed{\boldsymbol{-700\, e^{-0.04t}}}$	**Time-Path** $\boldsymbol{Q(t)} = \boxed{\boldsymbol{-700\, e^{-0.04t}}} + \mathbf{700}$
0	-700	0
1	-672.552	27.447392
50	-94.7346	605.26530
100	-12.8209	687.17905
150	-1.73512	698.26487
200	-0.23482	699.76517
250	-0.03177	699.96822
300	-0.00430	699.99569
350	-0.00058	699.99941
400	-0.00007	699.99992
450	-0.00001	699.99998
…	…	…

	Remarks	
	Negative deviation decays over time	Time-path **rises and converges** to equilibrium $(\boldsymbol{Q^*} = \mathbf{700})$
	As $\boldsymbol{t} \to \infty$, $\boldsymbol{Q_c} \to \mathbf{0}$	And when $\boldsymbol{Q_c} \to \mathbf{0}$, $(\boldsymbol{Q_t})\overrightarrow{Converges}(\boldsymbol{Q^*})$
	Graphically	

	
	The straight line shows the equilibrium value of Sales ($\boldsymbol{Q^*}$), while curve shows the deviation from equilibrium ($\boldsymbol{Q_c}$), and this deviation decreases over time and leads the time-path to convergence.

TOPIC 104: DYNAMICS OF ORE MINING USING DIFFERENTIAL EQUATIONS

The rate at which an ore is mined is given by the differential equation $\frac{dM}{dt} = \mathbf{0.02}\boldsymbol{M}$. Solve the differential equation, if the total reserves of ore in $\mathbf{1989}$ were $\mathbf{400}$ million tons.

Solution: Given the rate of change of Ore Mining $\left(\frac{dM}{dt} = \mathbf{0.02}\boldsymbol{M}\right)$, we can rearrange it to compare with the standard form of differential equations.

$$\frac{dM}{dt} = 0.02M$$

$$\frac{dM}{dt} - 0.02M = 0$$

$$\underbrace{\frac{dM}{dt}}_{\frac{dy}{dt}} - \underbrace{0.02}_{a}\,\underbrace{M}_{y} = \underbrace{0}_{b}$$

Here $\frac{dy}{dt} = \frac{dM}{dt}, y = M, a = -0.02, b = 0, y(0) = M(0) = 400$

Choice of Case of the Differential Equation:

$\boldsymbol{b} = \mathbf{0}$; **– Homogenous Case**.

$\boldsymbol{a} = -\mathbf{0.02}$; i.e. $\boldsymbol{a} \neq \mathbf{0}$ **– Homogenous Case with $\boldsymbol{a} \neq \mathbf{0}$.**

Therefore, using the definite Solution of time-path of $\boldsymbol{HC}$:

$$y(t) = \{y(0)\}e^{-at}$$

Emphasizing on the values to be substituted:

$$\underline{y}(t) = \left\{\underline{y(0)}\right\} e^{-\boxed{a}t}$$

Substituting values of $\boldsymbol{y}$ and $\boldsymbol{a}$, respectively:

$$\underline{M}(t) = \left\{\underline{M(0)}\right\} e^{\boxed{-}\boxed{(\boxed{-}0.02)}t}$$

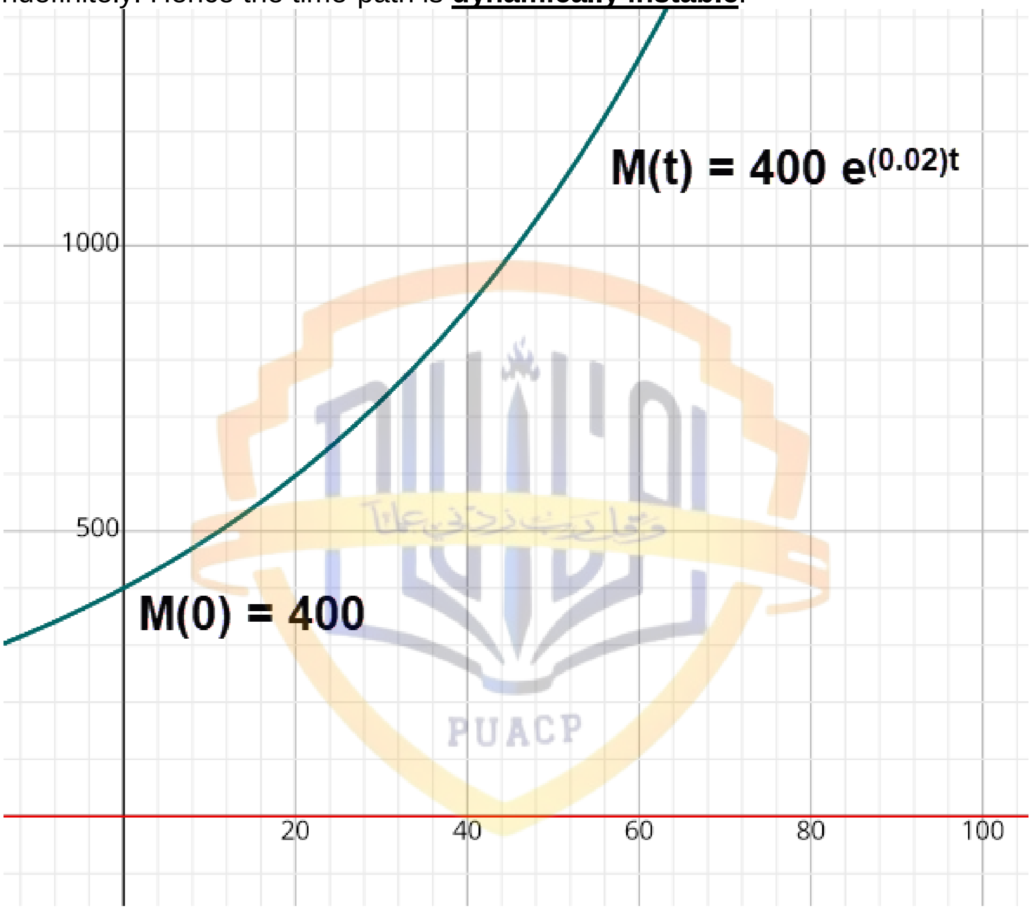
$$\boxed{M(t) = (400)e^{0.02t}}$$

Definite Time-path for Ore Mining in Dynamic Sense.

Dynamic Stability: For the dynamic stability of the time-path of First Order Differential Equation, the deviation expression should fade. In homogenous case ($\boldsymbol{HC}$), the particular solution is zero; $\boldsymbol{M_p} = \mathbf{0}$. Therefore, the convergence should take place towards zero ($\mathbf{0}$). In this case, we have a positive exponent $\boldsymbol{M}(\boldsymbol{t}) = (\mathbf{400})e^{\boxed{0.02}t}$which shall approach to infinity (∞), when $\boldsymbol{t} \to \infty$. In a differential equation, deviation component is measured by (M_c). Here, there is dynamic instability; $M_c \to \infty$ as $t \to \infty$.

$$\left[\underbrace{\{400\}}_{\textit{Deviation}}\ \underbrace{e^{\boxed{(0.02)}\,t}}_{\substack{\textit{Exponential}\\ \textit{Growth}\\ \textit{over time}}}\right] \to 0$$

Therefore, the time-path of Ore Mining shall **<u>diverge away from equilibrium</u>** as time is allowed to pass indefinitely. Hence the time-path is **<u>dynamically instable</u>**.



<u>TOPIC 105: DYNAMICS OF POLLUTANT USING DIFFERENTIAL EQUATIONS</u>

The amount of sulphur dioxide (in tons) present in the atmosphere is given by the equation $\frac{d(P)}{dt} = -0.2P$ (t is in hours). If 50 ton(s) is released into the atmosphere (i.e., $P = 50$ at $t = 0$), derive an expression for the amount of sulphur dioxide present at any time t.

Solution: Given the rate of change of Sulphur Dioxide (SO_2): $\frac{d(P)}{dt} = -0.2P$, we can rewrite it to compare it with the standard form of differential equation.

$$\frac{d(P)}{dt} = -0.2P$$

$$\frac{dP}{dt} + 0.2P = 0$$

$$\underbrace{\frac{dP}{dt}}_{\frac{dy}{dt}} + \underbrace{0.2}_{a}\underbrace{P}_{y} = \underbrace{0}_{b}$$

Here $\frac{dy}{dt} = \frac{dP}{dt}, y = P, a = 0.2, b = 0, y(0) = P(0) = 50$

Choice of Case of the Differential Equation:

$b = 0$; **– Homogenous Case**.

$a = 0.2$; i.e. $a \neq 0$ **– Homogenous Case with $a \neq 0$**.

Therefore, using the definite Solution of time-path of HC:

$$y(t) = \{y(0)\}e^{-at}$$

Emphasizing on the values to be substituted:

$$\underline{y}(t) = \left\{\underline{y(0)}\right\}e^{-\boxed{a}t}$$

Substituting values of y and a, respectively:

$$\underline{P}(t) = \left\{\underline{P(0)}\right\}e^{-\boxed{(0.2)}t}$$

$$P(t) = (\underline{50})e^{-0.2t}$$

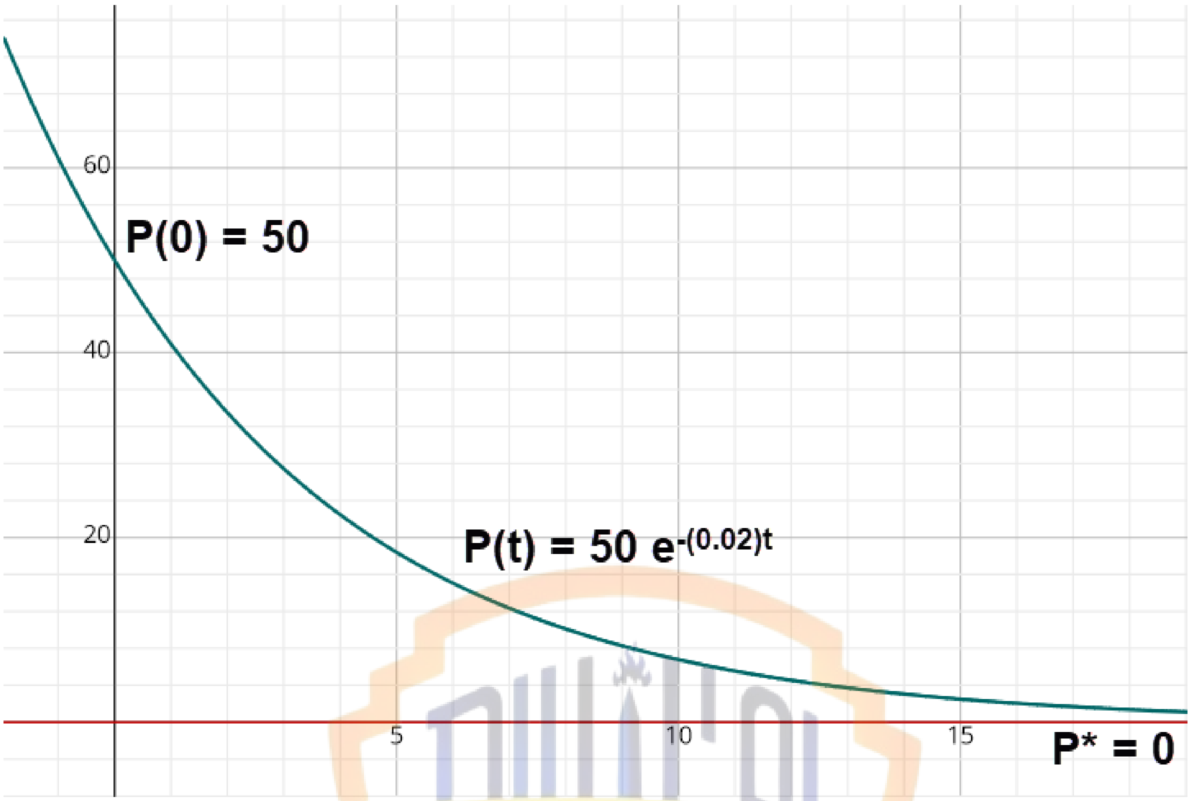
$$\boxed{P(t) = (50)e^{-0.2t}}$$

Definite Time-path for Government Expenditure in Dynamic Sense.

Dynamic Stability: For the dynamic stability of the time-path of First Order Differential Equation, the deviation expression should fade. In homogenous case (HC), the particular solution is zero; $P_p = 0$. Therefore, the convergence should take place towards zero (0). In this case, we have a negative exponent $P(t) = (50)e^{-\boxed{0.2}t}$ which shall approach to zero (0), when $t \to \infty$. In a differential equation, deviation component is measured by (P_c). Therefore, for dynamic stability; $P_c \to 0$ as $t \to \infty$.

$$\left[\underbrace{\{50\}}_{Deviation}\ \underbrace{e^{\boxed{-(0.2)}t}}_{\substack{Exponential\\ Decay\\ over\ time}}\right] \to 0$$

Therefore, the time-path of Pollutant shall **<u>converge towards equilibrium</u>** as time is allowed to pass indefinitely. Hence the time-path is **<u>dynamically stable</u>**.



TOPIC 106: DYNAMICS OF FISH STOCK USING DIFFERENTIAL EQUATIONS

The rate at which the fish stocks increase in an inland lake is given by the equation $\frac{dF}{dt} = \mathbf{0.12(80 - F)}$, where F is in thousands, t is in years. (a) Derive an expression for the total number of fish present at any time t given $F = 0$ at $t = 0$.
(b) Graph the total fish stock as a function of time from $t = 0$ to $t = 30$. Does the lake appear to have a maximum carrying capacity?
(c) Calculate the number of years taken for the fish stocks to reach 40000.

Solution: Given the rate of change of fish stock; $\frac{dF}{dt} = \mathbf{0.12(80 - F)}$, we can rearrange it to compare with the standard form of differential equation.

$$\frac{dF}{dt} = 0.12(80 - F)$$

$$\frac{dF}{dt} = 9.6 - 0.12F$$

$$\underbrace{\frac{dF}{dt}}_{\frac{dy}{dt}} + \underbrace{0.12}_{a}\underbrace{F}_{y} = \underbrace{9.6}_{b}$$

Here $\frac{dy}{dt} = \frac{dF}{dt}$, $y = F$, $a = 0.12$, $b = 9.6$, $y(0) = F(0) = 0$.

Choice of Case of the Differential Equation:

As $b = 9.6$, so $b \neq 0$: Non-Homogenous Case.

As $a = 0.12$, so $a \neq 0$: Non-Homogenous Case with $a \neq 0$.

Therefore, using the Definite Solution of time-path of $NHC_{a\neq 0}$:

$$y(t) = \left\{y(0) - \frac{b}{a}\right\}e^{-at} + \frac{b}{a}$$

Emphasizing on the values to be substituted:

$$\underline{y}(t) = \left\{\underline{y(0)} - \frac{\boxed{b}}{\boxed{a}}\right\} e^{-\boxed{a}t} + \frac{\boxed{b}}{\boxed{a}}$$

Substituting values of y, a, b and $y(0)$ respectively:

$$\underline{F}(t) = \left\{\underline{F(0)} - \frac{\boxed{9.6}}{\boxed{0.12}}\right\} e^{-\boxed{0.12}t} + \frac{\boxed{9.6}}{\boxed{0.12}}$$

$$F(t) = \left\{\underline{0} - \frac{9.6}{0.12}\right\} e^{-0.12t} + \frac{9.6}{0.12}$$

$$F(t) = (0 - 80)e^{-0.12t} + 80$$

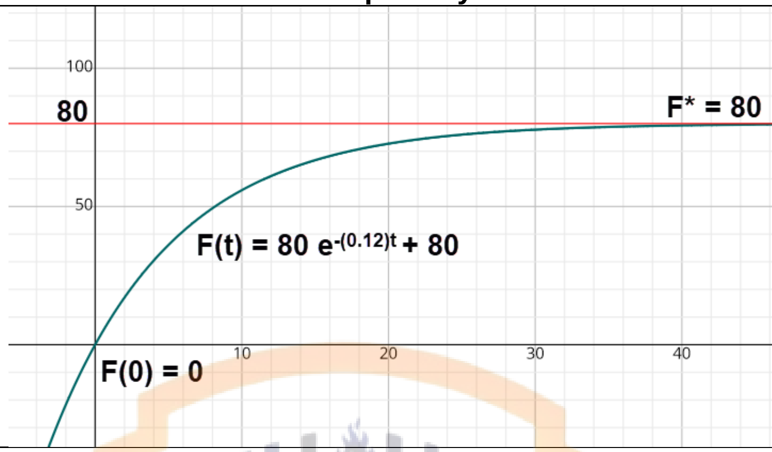
$$\boxed{F(t) = (-80)e^{-0.12t} + 80}$$

It is the definite Time-Path of Fish Stock.

$$F(t) = \underbrace{-80\, e^{-0.12t}}_{F_c} + \underbrace{80}_{F_p}$$

Dynamic Stability: The expression; $F(t) = \boxed{-80\, e^{-0.12t}} + 80$ has an exponential decay term in the box. **i.e.**:

	Complementary Function (Deviation)	**Time-Path**
t	$F_c = \boxed{-80\, e^{-0.12t}}$	$F(t) = \boxed{-80\, e^{-0.12t}} + 80$
0	-80	0
1	-70.95	9.0464
10	-24.10	55.904
20	-7.257	72.743
30	-2.186	77.814
40	-0.658	79.342
50	-0.198	79.802
60	-0.060	79.940
70	-0.018	79.982
80	-0.005	79.995
90	-0.002	79.998
…	…	…

	Remarks	
	Negative deviation decays over time	Time-path **rises and converges** to equilibrium $(\boldsymbol{F^* = 80})$
	As $\boldsymbol{t} \to \infty$, $\boldsymbol{F_c} \to \boldsymbol{0}$	And when $\boldsymbol{F_c} \to \boldsymbol{0}$, $(\boldsymbol{F_t})\overrightarrow{Converges}(\boldsymbol{F^*})$
	Graphically	
		
	The straight line shows the equilibrium value of Fish Stock $(\boldsymbol{F^*})$, while curve shows the deviation from equilibrium $(\boldsymbol{F_c})$, and this deviation decreases over time and leads the time-path to convergence.	